INCH-POUND
MIL-PRF-32383/7A
13 August 2026
SUPERSEDING
MIL-PRF-32383/7 w (1)
26 May 2023

PERFORMANCE SPECIFICATION SHEET
BATTERY, RECHARGEABLE, SEALED, SMALL TACTICAL UNIVERSAL BATTERY (STUB), BB-251XA/U and BB-252XA/U

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-32383A.

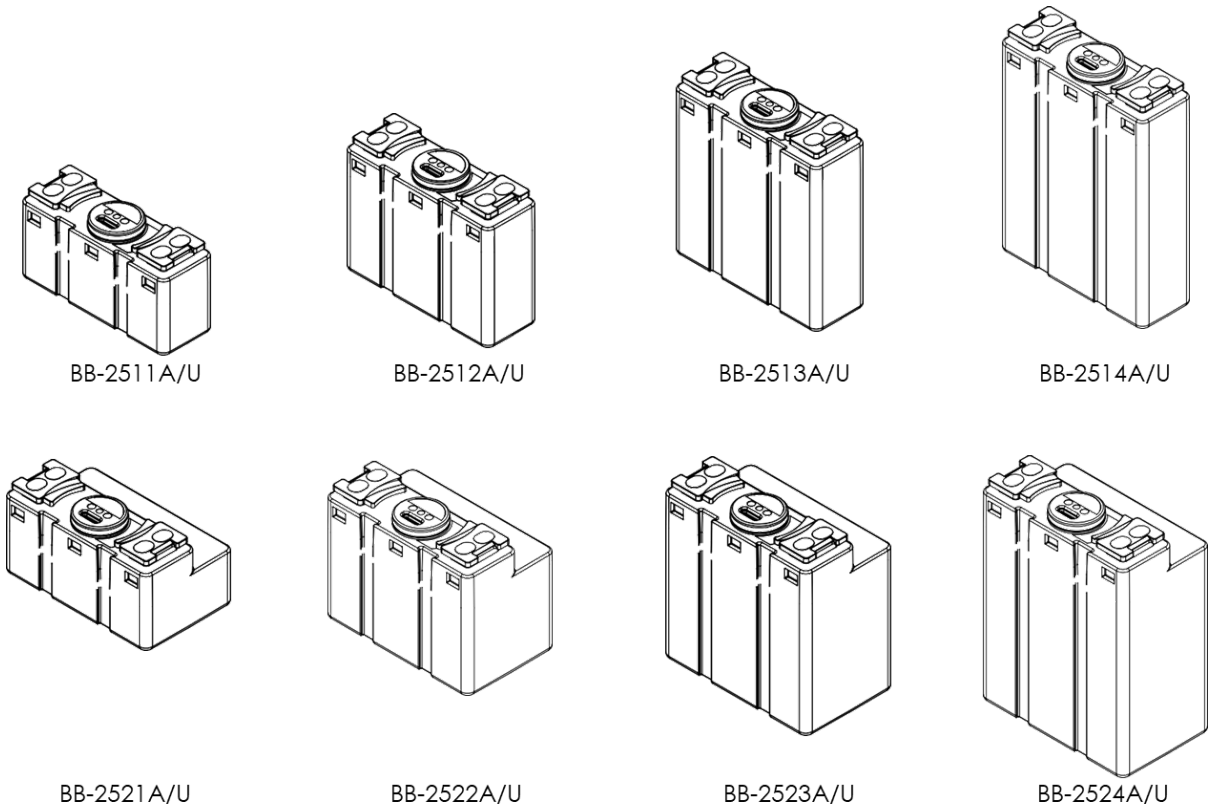


FIGURE 1. STUB configurations

FIGURE 1 NOTE:
Approximate sizes:

BB-2511A/U 2.85 in x 1.00 in x 1.50 in.
BB-2512A/U 2.85 in x 1.00 in x 2.25 in.
BB-2513A/U 2.85 in x 1.00 in x 3.00 in.
BB-2514A/U 2.85 in x 1.00 in x 3.75 in.
BB-2521A/U 2.85 in x 1.75 in x 1.50 in.
BB-2522A/U 2.85 in x 1.75 in x 2.25 in.
BB-2523A/U 2.85 in x 1.75 in x 3.00 in.
BB-2524A/U 2.85 in x 1.75 in x 3.75 in.

See FIGURE 4-14 for exact dimensions.

For clarity, MIL-PRF-32383A tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32383/7A tables are labeled with lower case Roman numerals (e.g. ix).
REQUIREMENTS

Distribution Statement A. Approved for public release: distribution is unlimited.

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For PIN (see 1.2 for PIN) designation see below

Battery	PIN	Battery	PIN
BB-2511A/U	M32383/7A-3-11	BB-2521A/U	M32383/7A-3-21
BB-2512A/U	M32383/7A-3-12	BB-2522A/U	M32383/7A-3-22
BB-2513A/U	M32383/7A-3-13	BB-2523A/U	M32383/7A-3-23
BB-2514A/U	M32383/7A-3-14	BB-2524A/U	M32383/7A-3-24

For the remainder of this document, the A/U will be omitted from the battery nomenclatures for brevity. For example, the BB-2511A/U will be listed as the BB-2511.

Specification requirements and test methods: The following requirements and test methods of MIL-PRF-32383A, identified therein as “when specified”, are applicable as indicated below:

TABLE i. *"When specified" requirements*

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Additional material requirements	*3.3.7*	*4.7.1.9*	*No*
Connector pull force	*3.4.4.1.1.1*	*Certification*	*No*
Connector insertions	*3.4.4.1.1.2*	*4.7.1.3*	*No*
Multilayer finish	*3.4.4.1.3*	*Certification*	*No*
Gold plating	*3.4.4.1.4*	*Certification*	*No*
USB power port connector(s)	3.4.4.1.5	4.7.2.17	Yes
USB Type-A connector(s)	*3.4.4.1.5.1*	*4.7.2.17*	*No*
USB Type-C connector(s)	3.4.4.1.5.2	4.7.2.17	Yes
Surface friction	3.4.5.6	4.7.1.8	Yes
Item unique identification (IUID)	*3.4.6.1.1*	*4.7.1.1*	*No*
Additional Markings	3.4.6.9	Certification	Yes
Additional physical characteristics requirements	*3.4.8*	*4.7.1.10*	*No*
Inrush current	3.5.5	4.7.2.12	Yes
Charge enable	*3.5.6.2*	*4.7.2.6*	*No*
Display, non-smart, always on	*3.5.8.1* *3.5.8.1.1*	*4.7.2.14*	*No*
Display, non-smart, user activated	3.5.8.1 3.5.8.1.2	4.7.2.14	Yes
Display, smart	*3.5.8.2*	*4.7.2.14*	*No*
SMBus	*3.5.9.1*	*4.7.2.15.1*	*No*
USB Power Delivery	3.5.9.2	4.7.2.15.2	Yes
Dual-role power device	3.5.9.2.1	4.7.2.15.2	Yes
Concurrent charge/discharge	*3.5.11*	*4.7.2.16*	*No*
Additional electrical requirements	3.5.12	4.7.2.18	Yes

TABLE i. *"When specified" requirements* – Continued

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Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Temperature shock high altitude opening (HAHO) simulation	*3.6.3*	*4.7.3.5.1*	*No*
Temperature shock high altitude low opening (HALO) simulation	*3.6.3*	*4.7.3.5.2*	*No*
Vibration, loose cargo transportation	3.6.5	4.7.3.7.1	Yes
Immersion, shallow	3.6.6	4.7.3.8.1	Yes
Immersion, deep	*3.6.6*	*4.7.3.8.2*	*No*
Transit drop, normal	*3.6.7*	*4.7.3.9.1*	*No*
Transit drop, severe	3.6.7	4.7.3.9.2	Yes
Drop while mated	*3.6.7*	*4.7.3.9.2*	*No*
Temperature soak	3.6.7	4.7.3.9.2b	Yes
Unmated drops	3.6.7	4.7.3.9.2c	Yes
Electrical tests	3.6.7	4.7.3.9.2d	Yes
Impact	*3.6.9*	*4.7.3.11*	*No*
Solar radiation	3.6.10	4.7.3.12	Yes
Salt fog	3.6.11	4.7.3.13	Yes
Electrostatic discharge (ESD)	*3.6.12*	*4.7.3.14*	*No*
Chemical resistance	3.6.13	4.7.3.15	Yes
Rapid decompression	3.6.14	4.7.3.16	Yes
Explosive atmosphere	3.6.15	4.7.3.17	Yes
Combined temperature, altitude, and humidity	3.6.16	4.7.3.18	Yes
Additional environmental requirements	*3.6.17*	*4.7.3.19*	*No*
Nail penetration, cell	*3.7.1.2*	*4.7.4.4*	*No*
Crush, cell	3.7.1.3	4.7.4.5	Yes
Inter-cell protective devices	*3.7.2.6*	*Certification*	*No*
Redundant safety protection	3.7.2.7	Certification	Yes
Projectile	3.7.2.8	4.7.4.11	Yes
Nail penetration, battery	3.7.2.9	4.7.4.12	Yes
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5	Yes
Weapons characterization	*3.7.2.11*	*4.7.4.13*	*No*
Additional safety requirements	*3.7.3*	*4.7.4.14*	*No*
Electromagnetic Interference	3.8	4.7.7	Yes
Additional transportation requirements	*3.9.1*	*4.7.8*	*No*
End item interoperability	3.10.2	4.7.6.2	Yes
Reliability	*3.11*	*4.7.2.4.1*	*No*
Additional test methods	*N/A*	*4.7.8*	*No*

Details: The following are details specific to M32383/7A-3-11, M32383/7A-3-12, M32383/7A-3-

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13, M32383/7A-3-14, M32383/7A-3-21, M32383/7A-3-22, M32383/7A-3-23 and M32383/7A-3-24 including those referenced in MIL-PRF-32383 as “specified”.

1. Non-Government publications (2.3).
   Add:
   Universal Serial Bus Type-C Cable and Connector Specification, Release 2.4

   Universal Serial Bus Power Delivery Specification, Revision 3.2, Version 1.1

   Delete:
   The USB Type-C Connectors and Cable Assemblies Compliance Document v 2.0

2. Certifications (3.2.1).
   Add: The following to TABLE II of MIL-PRF-32383A

REQUIREMENT	PARAGRAPH
USB certification marking	3.4.6.9.1
Display, non-smart, user activated (light intensity)	3.5.8.1.2
USB Power Delivery	3.5.9.2
Battery sleep mode	3.6.8.1

3. Dimensions and weight (3.4.3).

TABLE ii. *Nominal dimensions and weight*

	BB-2511	BB-2512	BB-2513	BB-2514	BB-2521	BB-2522	BB-2523	BB-2524
Weight	0.30 lb	0.40 lb	0.55 lb	0.65 lb	0.40 lb	0.65 lb	0.90 lb	1.20 lb
Width	2.85 in	2.85 in	2.85 in	2.85 in	2.85 in	2.85 in	2.85 in	2.85 in
Depth	1.00 in	1.00 in	1.00 in	1.00 in	1.75 in	1.75 in	1.75 in	1.75 in
Height	1.50 in	2.25 in	3.00 in	3.75 in	1.50 in	2.25 in	3.00 in	3.75 in

4. Connector (3.4.4.1.1).
   Connector: USB Type-C receptacle (see 2.3, The USB Type-C Connectors and Cable Assemblies Compliance Document Rev.2.1b) and the battery interface contacts (See FIGURE 2) engaged by these methods of attachment (see FIGURE 15).

   Mating connector: N/A
   Multilayer finish: N/A
   Gold plating: N/A

5. USB power port connector(s) (3.4.4.1.5).
   The battery shall contain one USB Type-C connector. The USB connector shall be located as shown in FIGURE 4.

6. USB Type-C connector(s) (3.4.4.1.5.2).
   a. See TABLE iv for Discharge rates
   c. The USB connector shall be located as shown in FIGURE 4.

Delete:

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b. A tethered protective cover

Delete:

7. Battery case vent (3.4.5.3).

Delete:

8. Terminal marking (3.4.6.4).

9. Marking (3.4.6).
For identification markings (see 3.4.6.1), a smaller font is allowed to include all required identification markings. The font size shall be maximized to the available space.

10. Identification marking (3.4.6.1).
Replace c. (Battery Voltage) V, (Full Capacity Rating) Ah / (Full Energy Rating) Wh with the following:

c. Contacts: 5 - {Max Voltage} V (FIXED/AVS/PPS), {Min BPDO Voltage} - {Max BPDO Voltage} V (BPDO), {PDP} W

USB-C: 5 - {Max Voltage} V (FIXED/AVS**/PPS), {Min BPDO Voltage} - {Max BPDO Voltage} V {PDP} W, (BPDO) {Maximum Allowable Power} W

*Information in curly brackets {} shall be filled in by the manufacturer.
**When applicable (see Source PDOs (3.5.12.12).)

Delete g. (MFG Warranty Details)

Add j. (CAGE code)

NOTE:
AVS = Adjustable Voltage Supply
BPDO = Battery Power Data Object
PDP = Power Delivery Power
PPS = Programmable Power Supply

11. Warning marking (3.4.6.2).
A smaller font is allowed to include all required warning markings. The font size shall be not less than 1.5 times the next largest font size used (see 3.4.6).

Delete:

12. Instructions / notes (3.4.6.5).

13. Simplified battery label (3.4.6.6).
See FIGURE 3 for location.

14. USB certification marking (3.4.6.9.1).
Batteries shall be marked with the USB Certified Charger Logo in the same area as the simplified label (see NOTES section for USB logo display requirements). See FIGURE 3 for location. Certification of USB marking is required.

15. Battery condition for shipping (3.4.7).

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Instruction card shall include:

a. Preferred charging method (Universal Battery Charger (UBC) 6130-01-659-7090, Advanced Battery Charger (ABC) 6130-01-660-3696)
b. Alternate charging method (any USB-PD certified charger)
c. Safety information
d. Storage information
e. Storage temperature
f. Operating temperature range
g. Charge temperature range
h. How to activate LEDs
i. STUB family with energy/dimensions/power delivery
j. STUB to STUB charging information
k. Description of terminals
l. If battery employs a shutdown mode, description of behavior and wake-up procedure.

The dust cap or cover shall cover contacts E1, E2, and E4 (see FIGURE 2).

16. Battery voltage (3.5.2).
Batteries shall operate within the *Power Data Object* (*PDO*) ranges of TABLE iii.

TABLE iii. *Fixed PDOs for capacity tests*

	BB-2511	BB-2512	BB-2513	BB-2514	BB-2521	BB-2522	BB-2523	BB-2524
Standard	5 V	9 V	9 V	15 V	9 V	15 V	15 V	20 V
High rate	5 V	9 V	15 V	15 V	9 V	15 V	20 V	20 V

When the battery is *Detached*, *VBUS* shall be ***vSafe0V***.

17. Capacity (3.5.3).
Cut-off voltage shall be in accordance with the USB-PD specification (see USB power delivery (3.5.9.2).

TABLE iv. *Standard/high-rate discharge*

	BB-2511	BB-2512	BB-2513	BB-2514	BB-2521	BB-2522	BB-2523	BB-2524
Standard	2.75 W	5.6 W	8.5 W	11.45 W	5.6 W	11.45 W	17.5 W	23.5 W
High rate	9 W	18 W	27 W	36 W	18 W	36 W	50 W	50 W

Each battery, discharged at the specified rate to cut-off voltage through the battery contacts unless otherwise specified, and shall have the following minimum capacity:

TABLE v. *Capacity table*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:08Z
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Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Energy (Wh) (via battery interface contacts)					
				BB-2511	BB-2512 & BB-2521	BB-2513	BB-2514 & BB-2522	BB-2523	BB-2524
Capacity Discharge	3.5.3	4.7.2.3	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Cycle life (224th cycle)	3.5.4	4.7.2.4	TABLE iii & TABLE iv	8.8	17.92	27.2	36.64	56	75.2
High rate discharge	3.5.3	4.7.2.5	TABLE iii & TABLE iv	8.8	17.92	27.2	36.64	56	75.2
Low temperature discharge	3.5.3	4.7.2.7	TABLE iii & TABLE iv	8.25	16.8	25.5	34.35	52.5	70.5
High temperature discharge	3.5.3	4.7.2.8	TABLE iii & TABLE iv	10.45	21.28	32.3	43.51	66.5	89.3
Charge acceptance	3.5.10.1	4.7.2.9	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
High temperature charge acceptance	3.5.10.1	4.7.2.9	TABLE iii & TABLE iv	9.35	19.04	28.9	38.93	59.5	79.9
Low temperature charge acceptance	3.5.10.1	4.7.2.9	TABLE iii & TABLE iv	9.35	19.04	28.9	38.93	59.5	79.9
Retention of charge	3.5.10.2	4.7.2.10	TABLE iii & TABLE iv	10.45	21.28	32.3	43.51	66.5	89.3
Pulse discharge	3.5.3	4.7.2.11	See 4.7.2.11	6.6	13.44	20.4	27.48	42	56.4
Extreme low temperature discharge	3.6.1	4.7.3.2	TABLE iii & TABLE iv	7.7	15.68	23.8	32.06	49	65.8
Extreme high temperature discharge	3.6.1	4.7.3.3	TABLE iii & TABLE iv	9.9	20.16	30.6	41.22	63	84.6
Temperature shock (post-test)	3.6.3	4.7.3.5	TABLE iii & TABLE iv	9.35	19.04	28.9	38.93	59.5	79.9

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TABLE v. *Capacity table* - Continued

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Energy (Wh) (via battery interface contacts)					
				BB-2511	BB-2512 & BB-2521	BB-2513	BB-2514 & BB-2522	BB-2523	BB-2524
Mechanical shock (post-test)	3.6.4	4.7.3.6	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Vibration (during test)	3.6.5	4.7.3.7	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Immersion	3.6.6	4.7.3.8	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Transit drop, severe	3.6.7	4.7.3.9.2	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Battery storage life	3.6.8	4.7.3.10	TABLE iii & TABLE iv	9.9	20.16	30.6	41.22	63	84.6
Solar Radiation	3.6.10	4.7.3.12	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Salt Fog	3.6.11	4.7.3.13	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Rapid decompression	3.6.14	4.7.3.16	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Explosive Atmosphere	3.6.15	4.7.3.17	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Combined temperature, altitude, and humidity	3.6.16	4.7.3.18	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Overcharge / electric leakage (post-test)	3.7.2.2	4.7.4.7	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94
Short circuit protection (post-test)	3.7.2.3	4.7.4.8	TABLE iii & TABLE iv	11	22.4	34	45.8	70	94

TABLE v. *Capacity table* - Continued

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Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Energy (Wh) (via battery interface contacts)					
				BB-2511	BB-2512 & BB-2521	BB-2513	BB-2514 & BB-2522	BB-2523	BB-2524
High temperature temporary cut off	3.7.2.4	4.7.4.9	TABLE iii & TABLE iv	10.45	21.28	32.3	43.51	66.5	89.3

18. Cycle life (3.5.4).
    Battery shall be capable of performing 224 cycles.

19. Inrush current (3.5.5).
    Not applicable for the USB Type-C interface.

    For the battery contacts interface described in FIGURE 2, the discharge profile shall be no less than listed in Table 6.10 Fixed Power Source Peak Current Capability in the USB Power Delivery Specification with bits 21…20 as 01b.

20. Charging interface (3.5.6.1).
    Battery shall have two charging interfaces; (1) interface described in FIGURE 2, and (2) USB Type-C receptacle.

21. Display, non-smart, user activated (3.5.8.1.2).
    Battery SOCI shall activate by user-activated button. If the button is pressed continuously, it shall be treated as one press. The button location and size shall be as shown in FIGURE 4.

22. USB power delivery (3.5.9.2).
    Replace with:
    When specified (see 3.1), batteries shall be compliant with USB PD Specification. When tested as specified in 4.7.2.15.2, batteries shall pass all USB PD Revision 3.2 Version 1.1 compliance tests or later. The battery shall be certified by the USB Integrators Forum (USB-IF). Certification of USB-IF certification is required.

    Add:
23. **Low SOC Test (3.5.12.1).**
    Batteries shall be able to complete three high power discharge pulses for 30 seconds at 20 % SOC.

Add:

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24. **Unstructured Vendor_Defined Messages (3.5.12.2).**
STUB shall support the following USB Power Delivery *Unstructured Vendor_Defined Messages* (*UVDM*s). Refer to USB Power Delivery Specification, Revision 3.2, Version 1.1, Section 6.4.4.1 Unstructured VDM for more information on *UVDM*s. The behavior defined in this section assumes that all rules related to *UVDM*s are being followed, e.g., an *Explicit Contract* is in place, STUB's *Port Partner* is the *DFP*, and STUB is the *UFP*. STUB shall not break any of the *UVDM* rules defined in the aforementioned section of the USB Power Delivery Specification.

a. VBUS Voltage and Current (0x00A1)
When STUB receives a *UVDM* containing the following *VDM Header*,

Bit(s)	Parameter	Value
B31…16	*Vendor ID (VID)*	0x38E7 (US Army Batteries)
B15	*VDM Type*	0 = *Unstructured VDM*
B14…0	Available for Vendor Use	0x00A1

STUB shall ignore any *Vendor Data Object* (*VDO*) that follows the *VDM Header* and return a *UVDM* containing an identical *VDM Header* and one *VDO* as follows:

Bits	Parameter	Units
B31…16	VBUS Voltage	mV
B15…0	VBUS Current	mA

a.i. VBUS Voltage
The VBUS Voltage field shall be a 16-bit unsigned integer with units of millivolts representing the instantaneous voltage of *VBUS* with respect to ground at the moment the incoming *UVDM* was received. The *VBUS* voltage is measured at the *VBUS* contacts or *VBUS* pins in the *USB Type-C* receptacle, whichever port is active. The measurement accuracy shall be ±3%.

a.ii. VBUS Current
The VBUS Current field shall be a 16-bit signed integer with units of milliamperes representing the instantaneous current of *VBUS* at the moment the incoming *UVDM* was received. A positive current represents the battery being charged. A negative current represents the battery being discharged. The *VBUS* current is measured at the *VBUS* contacts or *VBUS* pins in the *USB Type-C* receptacle, whichever port is active. The measurement accuracy shall be ±150 mA.

b. Cycle Count and State of Health (0x00A2)
When STUB receives a *UVDM* containing the following *VDM Header*,

Bit(s)	Parameter	Value
B31…16	*Vendor ID (VID)*	0x38E7 (US Army Batteries)
B15	*VDM Type*	0 = *Unstructured VDM*
B14…0	Available for Vendor Use	0x00A2

STUB shall ignore any *VDO* that follows the *VDM Header* and return a *UVDM* containing an

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identical *VDM Header* and one *VDO* as follows:

Bits	Parameter	Units
B31…16	Cycle Count	Cycles
B15…0	State of Health	%

b.i. Cycle Count
The Cycle Count field shall be a 16-bit unsigned integer representing the number of cycles the battery has experienced. A cycle is defined as an amount of discharge approximately equal to a STUB's design capacity.

b.ii. State of Health
The State of Health (SOH) field shall be a 16-bit unsigned integer representing the current state of health of the battery as a percentage.

SOH = (last full charge capacity/design capacity) x 100

c. Manufacture Date and Serial Number (0x00A3)

When STUB receives a *UVDM* containing the following *VDM Header*:

Bit(s)	Parameter	Value
B31…16	***Vendor ID (VID)***	0x38E7 (US Army Batteries)
B15	***VDM Type***	0 = *Unstructured VDM*
B14…0	Available for Vendor Use	0x00A3

STUB shall ignore any *VDO* that follows the *VDM Header* and return a *UVDM* containing an identical *VDM Header* and one *VDO* as follows:

Bits	Parameter	Units
B31…16	Manufacture Date	Year/Month/Day
B15…0	Serial Number	N/A

c.i. Manufacture Date
The Manufacture Date field shall be a 16-bit unsigned integer representing the date that the STUB was manufactured. The format of this field shall be as follows:

Bits	Parameter	Allowable Values
B31…25	Year-1980	0-127 (Years 1980-2107)
B24…21	Month	1-12
B20…16	Day	1-31

The Year field shall be the manufacture year minus 1980, which allows years up to 2107 to be encoded.

c.ii. Serial Number
The Serial Number field shall be a 16-bit unsigned integer representing a number that can be used to identify a particular STUB.

d. Other *UVDM*s

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When STUB receives a *UVDM* containing a *VDM Header* that does not exactly match any of the previous descriptions, it shall respond with a ***Not_Supported*** *Message* and take no further actions related to the *UVDM*.

Add:

25. **Ground impedance (3.5.12.3).**
Resistance shall be measured across the ground tabs and shall be not greater than 40 mΩ.

Add:

26. **Battery status message (3.5.12.4).**
Ref PD3.2v1.1 6.4.5.

The values for the ***Battery_Status*** *Message* shall be as follows for both the contacts and USB-C interfaces.

Battery Present Capacity shall report STUB's remaining energy in Watt-hours.

Fixed Battery 0 shall be the only valid *Battery* reference. ***Invalid Battery Reference*** shall be 0 for *Fixed Battery* 0 and shall be 1 for all other references.

Battery Present shall always be 1 for *Fixed Battery* 0.

Battery Charging Status shall accurately report whether STUB is currently charging, discharging, or idle.

Add:

27. **Extended Source capabilities (3.5.12.5).**
Ref PD3.2v1.1 6.5.1

The values for the ***Source_Capabilities_Extended*** *Message* shall be as follows for both the contacts and USB-C interfaces.

TABLE vi. *Source_Capabilities_Extended Message values*

Offset	Field	Value	Meaning of Value
0	***VID***	0x38E7	US Army Batteries
2	***PID***	0x2511 0x2512 0x2513 0x2514 0x2521 0x2522 0x2523 0x2524	Equal to STUB model number
4	***XID***	*	Manufacturer Specific
8	***FW Version***	*	Manufacturer Specific
9	***HW Version***	0x0A	Hardware version "A" for /7A

TABLE vi. *Source_Capabilities_Extended Message values* - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:08Z
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Offset	Field	Value	Meaning of Value
10	***Voltage Regulation***	0x00	• Load Step Slew Rate: 150 mA/µs • Load Step Magnitude: 25% *IoC*
11	***Holdup Time***	0x00	Not applicable; STUB does not depend on an *AC Supply* to operate as *Source*
12	***Compliance***	0x00	Not compliant with *LPS*, PS1, or PS2
13	***Touch Current***	0000 0010b	• Leakage current not < 65 µA rms when capability ≤ 30 W, or not < 100 µA rms when capability 30-100 W. • Does have a Ground pin. • Ground pin is not intended to be connected to protective earth.
14	***Peak Current1***	0x0000	Unused; see footnote 6 of TABLE viii and footnote 5 of TABLE ix
16	***Peak Current2***	0x0000	
18	***Peak Current3***	0x0000	
20	***Touch Temp***	0x00	Temperature conforms to ***[IEC 60950-1]***
21	***Source Inputs***	0000 0100b	• No external supply. • Internal *Battery* present.
22	***Number of Batteries/Battery Slots***	0x01	• 0 Hot Swappable Battery Slots. • 1 Fixed Battery.
23	***SPR Source PDP Rating***	*	Shall be equal to the PDP in TABLE viii for the contacts interface and TABLE ix for the USB-C interface.
24	***EPR Source PDP Rating***	0x00	0 W, *EPR* not supported

Add:

28. **Status message (3.5.12.6).**
Ref PD3.2v1.1 6.5.2

For both the contacts and USB-C interfaces, STUB shall accurately report all fields in the ***Status*** Message, including ***Internal Temp***, ***Event Flags***, ***Temperature Status***, and ***Power Status***. ***Temperature Status*** shall be supported.

Present Input shall be 0000 1000b to indicate Internal Power from *Battery*.

Present Battery Input shall be 0x01 to indicate *Fixed Battery* 0 is providing power.

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Check the source to verify that this is the current version before use.

Power State Change is optional and may be not supported.

Add:

29. **Battery capabilities message (3.5.12.7).**
Ref PD3.2v1.1 6.5.5

The values for the ***Battery_Capabilities*** *Message* shall be as follows for both the contacts and USB-C interfaces.

VID shall be 0x38E7 (US Army Batteries).

PID shall be equal to STUB model number.

Battery Design Capacity shall be not less than the values given in TABLE v for Capacity Discharge, shall be equal to the actual design capacity of the battery, and shall not be unknown.

Battery Last Full Charge Capacity shall be accurately reported and not unknown.

Fixed Battery 0 shall be the only valid *Battery* reference. ***Invalid Battery Reference*** shall be 0 for *Fixed Battery* 0 and shall be 1 for all other references.

Add:

30. **Manufacturer info (3.5.12.8).**
Ref PD3.2v1.1 6.5.7

The values for the ***Manufacturer_Info*** *Message* shall be as follows for both the contacts and USB-C interfaces.

VID shall be 0x38E7 (US Army Batteries).

PID shall be equal to STUB model number.

Manufacturer String shall be the same as or an abbreviated form of 3.4.6.1h.

Add:

31. **PPS status message (3.5.12.9).**
Ref PD3.2v1.1 6.5.10

The values for the ***PPS_Status*** *Message* shall be as follows for both the contacts and USB-C interfaces.

Output Voltage shall be supported.

Output Current may be supported.

PTF and ***OMF*** shall be supported and accurately reported.

Add:

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32. **Extended Sink capabilities (3.5.12.10).**
STUB shall support not less than the following extended *Sink* capabilities. When tested as specified in Extended Sink capabilities (4.7.2.18.10), the STUB shall accurately return the ***Sink_Capabilities_Extended*** *Message* (ref. PD3.2 V1.1 6.5.13).

TABLE vii. *Sink_Capabilities_Extended Message values*

Offset	Field	Value	Meaning of Value
0	***VID***	0x38E7	US Army Batteries
2	***PID***	0x2511 0x2512 0x2513 0x2514 0x2521 0x2522 0x2523 0x2524	Equal to STUB model number.
4	***XID***	*	Manufacturer Specific
8	***FW Version***	*	Manufacturer Specific
9	***HW Version***	0x0A	Hardware version “A” for /7A
10	***SKEDB Version***	0x01	Version 1.0
11	***Load Step***	0x00	150 mA/µS
12	***Sink Load Characteristics***	0x8000	• Percent overload = 0% • Overload period = N/A • Duty cycle = N/A • Can tolerate V_{BUS} voltage droop
14	***Compliance***	0x00	• Does not require *LPS Source* • Does not require PS1 *Source* • Does not require PS2 *Source*
15	***Touch Temp***	0x00	Not applicable
16	***Battery Info***	0x01	• 1 *Fixed Battery* • 0 Hot Swappable *Battery Slots*
17[1, 2]	***Sink Modes***	0x3F	• *PPS* charging supported • V_{BUS} powered • *AC Supply* powered • *Battery* powered • *Battery* essentially unlimited • *AVS* Support as Sink

TABLE vii. *Sink_Capabilities_Extended Message values* - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:08Z
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Offset	Field	Value	Meaning of Value
18[3]	***SPR Sink Minimum PDP***	0x00	0 W
19[4, 5]	***SPR Sink Operational PDP***	0x01	• Contacts: 0.5 W • USB-C: 1 W
20[6]	***SPR Sink Maximum PDP***	0x07 0x0E 0x15 0x1C 0x0E 0x1C 0x2A 0x38	BB-2511A: 7 W BB-2512A: 14 W BB-2513A: 21 W BB-2514A: 28 W BB-2521A: 14 W BB-2522A: 28 W BB-2523A: 42 W BB-2524A: 56 W
21	***EPR Sink Minimum PDP***	0x00	Not *EPR Capable*
22	***EPR Sink Operational PDP***	0x00	Not *EPR Capable*
23	***EPR Sink Maximum PDP***	0x00	Not *EPR Capable*

[1] If STUB's internal cell voltage is such that it cannot accept the *Source PPS* voltage range, byte 17 may clear bit 0 to 0 (i.e. 0011 1110b)

[2] USB-C interface only: If STUB has *PDP* less than 27 W, it shall not support *AVS* per PD3.2 V1.1 10.2.2 and offset byte 17 shall clear bit 5 to 0 (i.e. 0001 1111b)

[3] Shall be 0 W per 6.5.13.13 since STUB is a battery (i.e. self-powered)

[4] STUB shall charge and shall not indicate a ***Capability Mismatch*** for *Source PDP* rating of not less than 0.5 W (for the contacts interface) or 1 W (for the USB-C interface) per PD3.2 V1.1 10.3.2

[5] Contacts interface only: If a cell heater is used to meet MIL-PRF-32383A 3.5.10.1, and the heater is to be activated, byte 19 shall equal the minimum power necessary to operate the heater. If *Source* advertises insufficient *PDP*, STUB shall indicate a ***Capability Mismatch***.

[6] SPR Sink Maximum PDP shall be not less than the listed values for each STUB

Add:

33. **Ports Isolation (3.5.12.11).**
STUB shall have isolated ports between the *USB Type-C* connector and the battery contacts. When tested as specified in Ports Isolation (4.7.2.18.11), when any and/or all *CC* pins or contacts of one port are shorted to ground, the other port shall function normally. When voltage is present on *VBUS* on one port when an *Explicit Contract* is negotiated, the voltage of *VBUS* on the other port shall be 0 ± 100 mV with respect to ground.

Add:

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34. **Source PDOs (3.5.12.12).**
When tested as specified in Source PDOs (4.7.2.18.12), STUB shall have the following *PDO*s in its ***Source_Capabilities*** *Message*.

TABLE viii. *Source PDOs for Contacts and USB-C Interfaces*

	Source (Contacts)[1, 2]	Fixed PDO[3, 4] 5 V	9 V	15 V	20 V	Battery PDO (BPDO)[5]	Augmented PDO (APDO) SPR AVS[6]	SPR PPS[7]	PDP[8]
STUB	**BB-2511A**	3.60 A	2.00 A	1.20 A	0.90 A	2.50 - 4.20 V / 18.00 W	9 - 15 V / 1.20 A 15 - 20 V / 0.90 A	5 - 11 V / 2.00 A	18 W
	BB-2512A	5.00 A	4.00 A	2.40 A	1.80 A	5.00 - 8.40 V / 36.00 W	9 - 15 V / 2.40 A 15 - 20 V / 1.80 A	5 - 11 V / 4.00 A	36 W
	BB-2513A	5.00 A	5.00 A	3.60 A	2.70 A	7.50 - 12.60 V / 54.00 W	9 - 15 V / 3.60 A 15 - 20 V / 2.70 A	5 - 16 V / 3.60 A	54 W
	BB-2514A	5.00 A	5.00 A	4.80 A	3.60 A	10.00 - 16.80 V / 72.00 W	9 - 15 V / 4.80 A 15 - 20 V / 3.60 A	5 - 21 V / 3.60 A	72 W
	BB-2521A	5.00 A	4.00 A	2.40 A	1.80 A	5.00 - 8.40 V / 36.00 W	9 - 15 V / 2.40 A 15 - 20 V / 1.80 A	5 - 11 V / 4.00 A	36 W
	BB-2522A	5.00 A	5.00 A	4.80 A	3.60 A	10.00 - 16.80 V / 72.00 W	9 - 15 V / 4.80 A 15 - 20 V / 3.60 A	5 - 21 V / 3.60 A	72 W
	BB-2523A	5.00 A	5.00 A	5.00 A	5.00 A	7.50 - 12.60 V / 60.00 W	9 - 15 V / 5.00 A 15 - 20 V / 5.00 A	5 - 21 V / 5.00 A	100 W
	BB-2524A	5.00 A	5.00 A	5.00 A	5.00 A	10.00 - 16.80 V / 80.00 W	9 - 15 V / 5.00 A 15 - 20 V / 5.00 A	5 - 21 V / 5.00 A	100 W

	Source (USB-C)[2, 9]	Fixed PDO[3, 10] 5 V	9 V	15 V	20 V	Battery PDO (BPDO)	Augmented PDO (APDO) SPR AVS[6]	SPR PPS[7]	PDP[8]
STUB	**BB-2511A**	3.60 A	2.00 A	-	-	2.50 - 4.20 V / 12.50 W	- -	5 - 11 V / 2.00 A	18 W
	BB-2512A	5.00 A	4.00 A	2.40 A	-	5.00 - 8.40 V / 25.00 W	9 - 15 V / 2.40 A -	5 - 11 V / 4.00 A	36 W
	BB-2513A	5.00 A	5.00 A	3.60 A	2.70 A	7.50 - 12.60 V / 37.50 W	9 - 15 V / 3.60 A 15 - 20 V / 2.70 A	5 - 16 V / 3.60 A	54 W
	BB-2514A	5.00 A	5.00 A	4.80 A	3.60 A	10.00 - 16.80 V / 50.00 W	9 - 15 V / 4.80 A 15 - 20 V / 3.60 A	5 - 21 V / 3.60 A	72 W
	BB-2521A	5.00 A	4.00 A	2.40 A	-	5.00 - 8.40 V / 25.00 W	9 - 15 V / 2.40 A -	5 - 11 V / 4.00 A	36 W
	BB-2522A	5.00 A	5.00 A	4.80 A	3.60 A	10.00 - 16.80 V / 50.00 W	9 - 15 V / 4.80 A 15 - 20 V / 3.60 A	5 - 21 V / 3.60 A	72 W
	BB-2523A	5.00 A	5.00 A	5.00 A	5.00 A	7.50 - 12.60 V / 37.50 W	9 - 15 V / 5.00 A 15 - 20 V / 5.00 A	5 - 21 V / 5.00 A	100 W
	BB-2524A	5.00 A	5.00 A	5.00 A	5.00 A	10.00 - 16.80 V / 50.00 W	9 - 15 V / 5.00 A 15 - 20 V / 5.00 A	5 - 21 V / 5.00 A	100 W

[1] If >4A of current is drawn from battery contacts, both *VBUS* contacts shall be used by the *Sink*

[2] *PDO*s shall be advertised in the order presented (left to right) in this table per PD3.2 V1.1 6.4.1.4

[3] 12V *Fixed PDO* from MIL-PRF-32383/7 deprecated per PD3.2 V1.1 10.2.3.1

[4] Bits 21…20 of the *PDO* shall be 01b for the ***Peak Current*** field (ref. PD3.2 V1.1 6.4.1.2.1.8, Table 6.10, and 7.1.11). VBUS Droop is allowed (ref. PD3.2 V1.1 7.1.11).

[5] *Source BPDO* shall allow for negative current and begin charging STUB at not greater than the value of Offset Byte 19 of the SKEDB and accept not less than the value of Offset Byte 20 of the SKEDB

[6] Bits 27…26 of the *SPR AVS APDO* shall be the same as bits 21…20 of the *Fixed PDO* (ref. PD3.2 V1.1 6.4.1.2.4.2.1). *VBUS* Droop is allowed (ref. PD3.2 V1.1 7.1.11).

[7] ***PPS Power Limited*** bit shall be set to 1 (ref. PD3.2 V1.1 6.4.1.2.4.1.1)

[8] *PDP* limited to 100W per *SPR Mode* definition in PD3.2 V1.1 1.2

[9] *PDO*s listed assume use of a 5A capable cable.
Otherwise, all advertised values of current otherwise greater than 3A shall be equal to 3A.

[10] Bits 21…20 of the *PDO* shall be 00b for the ***Peak Current*** field (ref. PD3.2 V1.1 6.4.1.2.1.8 and Table 6.10)

Add:

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35. **Disambiguation of Power Roles (3.5.12.13).**
STUB shall have two methods that enable the user to designate STUB as a *Source*:

1) If the user presses the button on STUB before connecting it to another DRP device, the STUB shall temporarily act as a *Source*-only device for 10 seconds.

2. a. i.) If the user presses the button on STUB after it is connected to another DRP device and the STUB is *Sink*, the STUB shall send a ***PR_Swap*** *Message*. If STUB receives an ***Accept*** *Message* in response, STUB shall perform a *Power Role Swap* per the USB Power Delivery specification.

2. a. ii.) STUB shall be capable of sending an ***Accept*** *Message* if it is *Source* and receives a ***PR_Swap*** *Message*. STUB shall perform a *Power Role Swap* after sending an ***Accept*** *Message* per the USB Power Delivery specification.

2. b.) If the user presses the button on STUB after it is connected to another DRP device and the STUB is *Source*, STUB shall activate its state-of-charge indicator consistent with normal operation of the button. STUB shall not do anything else in response to the button press.

Replace 3.6.8.1 Battery sleep mode with the following:

36. Battery sleep mode (3.6.8.1).
The battery shall not enter sleep mode within 48 hours of the last activity (i.e. button press, charge current, or battery load). Certification of the 48-hour delay is required.

37. Battery wake up (3.6.8.2).
The battery shall wake up from sleep mode after not greater than two seconds following any of the following conditions:
    a. The battery's button is pressed.
    b. The battery is connected to a charger via the USB Type-C receptacle.
    c. The battery is connected to a charger via its battery contacts.

The battery shall support all three of the above conditions.

38. High temperature temporary cut off devices (3.7.2.4).
Not less than two devices

39. High temperature permanent cut off devices (3.7.2.5).
Not less than one device

40. Redundant safety protection (3.7.2.7).
Batteries shall have a battery management system (BMS) electrically protecting all the terminals and external connections. The BMS shall include not less than:
    a. two current control devices in series (e.g., Field-Effect Transistor (FET))
    b. one resettable current fuse (e.g., Positive Temperature Coefficient (PTC))
    c. one permanent current fuse

All three tiers of protection shall have had to fail prior to allowing any potential uncontrolled power flow to or from the battery cells that would cause an unsafe condition from any signal or power pins on any connector ports.
NOTE:

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Components used to satisfy 3.7.2.4 and 3.7.2.5 may also be used to satisfy redundant safety protection requirements.

41. End item interoperability (3.10.2).
End item(s) will be specified prior to First Article Testing (FAT).

VERIFICATION

42. First article inspection (4.4).
Replace e, f, and g with the following:

e. Group 2 (GR2) – Electromagnetic interference and nail penetration tests, shall contain three samples
f. Group 3 (GR3) – Interoperability tests, shall contain six samples
g. Group 4 (GR4) – Lithium safety tests, shall contain 18 samples

Replace TABLE X with the following:

TABLE X. *First article tests, complete batteries*

Inspection/ Test	Req't Paragraph	Test Method Paragraph	Samples or Group (GR)												
			GR 1										GR 2	GR 3	GR 4
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(3 ea)	(6 ea)	(18 ea)
Visual and mechanical	3.4.1	4.7.1.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Battery open circuit voltage	3.5.2	4.7.2.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Ground Impedance	3.5.12.3	4.7.2.18.3	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Insulation resistance	3.7.2.1	4.7.4.6	✓	✓	✓	✓	✓	✓	✓	✓					
Altitude	3.6.2	4.7.3.4	✓	✓	✓	✓	✓	✓	✓	✓					
Charge acceptance	3.5.10.1	4.7.2.9								✓					
Capacity discharge (initial)	3.5.3	4.7.2.3.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Cycle life	3.5.4	4.7.2.4					✓								
Battery storage life	3.6.8	4.7.3.10						✓							
Overcharge / electrical leakage	3.7.2.2	4.7.4.7							✓						

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TABLE X. *First article tests, complete batteries* - Continued

Inspection/ Test	Req't Paragraph	Test Method Paragraph	Samples or Group (GR)												
			GR 1										GR 2	GR 3	GR 4
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(3 ea)	(6 ea)	(18 ea)
Low temperature discharge	3.6.1	4.7.2.7							✓						
High temperature discharge	3.6.1	4.7.2.8							✓						
Projectile	3.7.2.8	4.7.4.11							✓						
High rate discharge	3.5.3	4.7.2.5	✓		✓										
Retention of charge	3.5.10.2	4.7.2.10	✓	✓	✓	✓									
Pulse discharge	3.5.3	4.7.2.11	✓		✓										
Inrush current	3.5.5	4.7.2.12	✓		✓										
Low SOC	3.5.12.1	4.7.2.18.1	✓		✓										
Temperature shock	3.6.3	4.7.3.5	✓	✓	✓	✓									
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓									
Vibration (during test)	3.6.5	4.7.3.7	✓	✓	✓	✓									
Vibration (loose cargo)	3.6.5	4.7.3.7.1	✓	✓	✓	✓									
Chemical resistance	3.6.13	4.7.3.15									✓				
Flat terminal strength	3.4.4.1.2	4.7.1.4	✓		✓										
USB power port connector(s)	3.4.4.1.5	4.7.2.17	✓	✓	✓	✓									
Ports isolation	3.5.12.11	4.7.2.18.11	✓	✓	✓	✓									
Immersion, shallow	3.6.6	4.7.3.8.1								✓		✓		✓	
Transit drop, severe	3.6.7	4.7.3.9.2								✓		✓		✓	
Surface friction	3.4.5.6	4.7.1.8		✓	✓										
Salt fog	3.6.11	4.7.3.13									✓				
Solar Radiation	3.6.10	4.7.3.12									✓				
Rapid decompression	3.6.14	4.7.3.16	✓	✓	✓	✓									

TABLE X. *First article tests, complete batteries* - Continued

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Inspection/ Test	Req't Paragraph	Test Method Paragraph	Samples or Group (GR)												
			GR 1										GR 2	GR 3	GR 4
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(3 ea)	(6 ea)	(18 ea)
Explosive atmosphere	3.6.15	4.7.3.17	✓			✓									
Combined temperature, humidity, and altitude	3.6.16	4.7.3.18									✓				
Immersion, shallow (post drop)	3.6.6	4.7.3.8.1								✓		✓		✓	
State of charge indicator	3.5.8	4.7.2.14									✓				
Full capacity discharge	3.5.3	4.7.2.3	✓	✓	✓	✓									
Extreme low temp. discharge	3.6.1	4.7.3.2		✓	✓										
Extreme high temp. discharge	3.6.1	4.7.3.3		✓	✓										
Short circuit protection	3.7.2.3	4.7.4.8											✓		
High temp. temporary cut-off	3.7.2.4	4.7.4.9								✓				✓	
High temp. permanent cut-off	3.7.2.5	4.7.4.10		✓	✓										
Electromagnetic interference	3.8	4.7.7											✓		
End item interoperability	3.10.2	4.7.6.2												✓	
Charger interoperability	3.10.1	4.7.6.1												✓	
Nail penetration, battery	3.7.2.9	4.7.4.12											✓		
Battery status message	3.5.12.4	4.7.2.18.4	✓			✓									
Extended Source capabilities	3.5.12.5	4.7.2.18.5	✓			✓									
Status message	3.5.12.6	4.7.2.18.6	✓			✓									

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TABLE X. *First article tests, complete batteries* – Continued

Inspection/ Test	Req't Paragraph	Test Method Paragraph	Samples or Group (GR)												
			GR 1										GR 2	GR 3	GR 4
			1	2	3	4	5, 6	7, 8	9, 10	11, 12	13, 14	15, 16	(3 ea)	(6 ea)	(18 ea)
Battery capabilities message	3.5.12.7	4.7.2.18.7	✓			✓									
Manufacturer info	3.5.12.8	4.7.2.18.8	✓			✓									
PPS status message	3.5.12.9	4.7.2.18.9	✓			✓									
Extended Sink capabilities	3.5.12.10	4.7.2.18.10	✓			✓									
Source PDOs	3.5.12.12	4.7.2.18.12	✓			✓									
Disambiguation of power roles	3.5.12.13	4.7.2.18.13	✓			✓									
Unstructured Vendor_Defined Messages	3.5.12.2	4.7.2.18.2	✓			✓									
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5													✓

43. Alternate charging (4.6.3).
    1. Use of 4.6.3 permitted
    2. *USB-certified PD Charger*
    3. *USB-certified PPS Charger*

44. Charge acceptance (3.5.10.1).
    c. Use of 4.6.3 is permitted.

    Test shall be repeated using the USB Type-C interface and battery contacts for charging.

45. Battery open circuit voltage (4.7.2.2).
    The OCV test shall consist of the standard PDO (see TABLE iii) and ***vSafe0V*** (Battery voltage (3.5.2).) measurements. Measurements shall be taken at both the contacts and the USB-C interface.

46. Capacity discharge (4.7.2.3).
    c. Discharge at the specified PDOs in TABLE iii at the rates specified in TABLE iv.

47. Cycle life test (4.7.2.4).

Delete: part b.

Replace c. with the following:

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c. Repeat step a above until the total number of required cycles is obtained (see 3.1). Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Perform the electric leakage test of 4.7.4.7e. Allow batteries to rest for not less than five minutes and not greater than 16 hours. Discharge at the rate specified to the specified cutoff voltage. Record battery capacity.

Battery	Power (W)	PDO (V)
BB-2511	3.67	5
BB-2512	7.47	9
BB-2513	11.33	9
BB-2514	15.27	15
BB-2521	7.47	9
BB-2522	15.27	15
BB-2523	23.33	20
BB-2524	31.33	20

48. Pulse discharge (4.7.2.11).

d. Discharge with the battery at specified pulse power and duration. First pulse shall be run for 144 s, followed by 18 s for second pulse and 18 s for third pulse. Continue the pulse sequence until the battery reaches 0 % SOC or until a fault event (i.e., high temperature) is triggered. Ramp up or down time between successive pulses shall be not greater than 2 seconds and shall abide by USB-PD specifications at all times.

Battery	Power (W)	PDO (V)
BB-2511	2.75 / 4.12 / 18	5
BB-2512	5.6 / 8.4 / 36	9
BB-2513	8.5 / 12.75 / 54	15
BB-2514	11.45 / 17.17 / 72	15
BB-2521	5.6 / 8.4 / 36	9
BB-2522	11.45 / 17.17 / 72	15
BB-2523	17.5 / 26.25 / 100	20
BB-2524	23.5 / 35.25 / 100	20

49. Inrush current (4.7.2.12).

d. Each PDO where the ***Peak Current*** field is applicable (i.e., *Fixed Supply PDO*s and *AVS APDO*s) shall be tested. For *AVS APDOs*, the midpoint voltage of the range(s) shall be used. Each of the three Overload *Capabilities* shall be tested. Each Overload *Capability* shall be tested for three pulses. Each test shall begin at the low current value. There shall be one more low-current period following the final peak current period.

IoC shall be equal to the ***Maximum Current*** of a given PDO.

50. USB power port connector(s) (4.7.2.17).

b. Certify as part of USB-IF certification (see USB power delivery (3.5.9.2).)
d. See FIGURE 2
e. Fulfilled by Ports Isolation (4.7.2.18.11).

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Add:

51. **Low SOC Test (4.7.2.18.1).**

The following test shall be performed.

a. Discharge the batteries completely to 0 % SOC at specified standard discharge rate.
b. Charge the batteries to 20 % SOC in accordance with 4.6; use of 4.6.3 is permitted.
c. Disconnect the batteries for a minimum of 10 minutes.
d. Discharge the batteries at max power (PDP) specified for the battery for 30 seconds. Ramp up time to reach PDP shall be not greater than 2 seconds and shall abide by USB-PD specifications at all times.
e. Repeat c and d for a minimum of 3 times.
f. After test, battery shall meet the requirements of Low SOC Test (3.5.12.1).

Add:

52. **Unstructured Vendor Defined Messages (4.7.2.18.2).**

The following test shall be performed. USB tester shall be set up to send to STUB each defined *UVDM* containing the specified *VDM Header* and no *VDO*s. Check that STUB returns a *UVDM* with the correct information. Then, repeat the test with the USB tester sending STUB each specified *UVDM* with any arbitrary *VDO*(s) following the *VDM Header*. Check that STUB still returns a *UVDM* with the correct information. Finally, the USB tester shall send to STUB a *UVDM* that does not match any of the specified *UVDM*s. Check that STUB returns a ***Not_Supported*** *Message* and did not take any further actions related to the *UVDM*s. Testing shall be performed through the battery contacts and USB Type-C connector. The battery shall meet the requirements of Unstructured Vendor_Defined Messages (3.5.12.2).

Add:

53. **Ground impedance (4.7.2.18.3).**

The following test shall be performed. Record impedance across the two ground tabs of battery, see FIGURE 6. Battery shall meet requirements of Ground impedance (3.5.12.3).

Add:

54. **Battery status message (4.7.2.18.4).**

The following test shall be performed. A device shall be set up to read STUB's ***Battery_Status*** *Message* by sending the ***Get_Battery_Status*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of Battery status message (3.5.12.4).

Add:

55. **Extended Source capabilities (4.7.2.18.5).**

The following test shall be performed. A device shall be set up to read STUB's ***Source_Capabilities*** *Message* and ***Source_Capabilities_Extended*** *Message* by sending the ***Get_Source_Cap*** *Message* ***Get_Source_Cap_Extended*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of Extended Source capabilities (3.5.12.5).

Add:

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56. **Status message (4.7.2.18.6).**
The following test shall be performed. A device shall be set up to read STUB's ***Status*** *Message* by sending the ***Get_Status*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of Status message (3.5.12.6).

Add:

57. **Battery capabilities message (4.7.2.18.7).**
The following test shall be performed. A device shall be set up to read STUB's ***Battery_Capabilities*** *Message* by sending the ***Get_Battery_Cap*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of Battery capabilities message (3.5.12.7).

Add:

58. **Manufacturer info (4.7.2.18.8).**
The following test shall be performed. A device shall be set up to read STUB's ***Manufacturer_Info*** *Message* by sending the ***Get_Manufacturer_Info*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of Manufacturer info (3.5.12.8).

Add:

59. **PPS status message (4.7.2.18.9).**
The following test shall be performed. A device shall be set up to read STUB's ***PPS_Status*** *Message* by sending the ***Get_PPS_Status*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of PPS status message (3.5.12.9).

Add:

60. **Extended Sink capabilities (4.7.2.18.10).**
The following test shall be performed. A device shall be set up to read STUB's ***Sink_Capabilities*** *Message* and ***Sink_Capabilities_Extended*** *Message* by sending the ***Get_Sink_Cap*** *Message* ***Get_Sink_Cap_Extended*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector. Battery shall meet the requirements of Extended Sink capabilities (3.5.12.10).

Add:

61. **Ports Isolation (4.7.2.18.11).**
The following test shall be performed.
    a. Verify that the ports are isolated by shorting the *CC* contact to the ground tab and then connecting any functional USB Power Delivery device via the USB Type-C receptacle. Check that an *Explicit Contract* was successfully made, i.e., STUB is charging if a charger was connected or the device is powered on if a *Sink* was connected. Verify the voltage of the *VBUS* contact and USB-C connector pins. Disconnect the device from STUB.
    b. Then, short STUB's CC1 pin to ground in the USB-C connector. Connect a functional USB Power Delivery device via the contacts and check if a contract was successfully made. Verify the voltage of the *VBUS* contact and USB-C connector pins.
    c. Then, short STUB's CC2 pin to ground in the USB-C connector. Connect a functional USB Power Delivery device via the contacts and check if a contract was successfully made. Verify the voltage of *VBUS* in the USB Type-C

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connector.

d. Finally, short STUB's CC1 and CC2 pins to ground in the USB-C connector. Connect a functional USB Power Delivery device via the contacts and check if a contract was successfully made. Verify the voltage of the *VBUS* contact and USB-C connector pins.

Battery shall meet the requirements of Ports Isolation (3.5.12.11).

Add:

62. **Source PDOs (4.7.2.18.12).**

The following test shall be performed. A USB tester shall be set up to read STUB's ***Source_Capabilities*** *Message* by sending the ***Get_Source_Cap*** *Message*. Testing shall be performed through the battery contacts and USB Type-C connector.

To test that the BPDO allows negative current, STUB shall start the test with a state of charge of not greater than 90%. STUB shall output the BPDO. A USB PD tester, device, or trigger may be necessary to get STUB to output the BPDO. A power supply shall be connected to the VBUS and GND battery contacts. The power supply shall output a constant voltage equal to the maximum voltage of the BPDO and the current shall be monitored. Battery shall meet the requirements of Source PDOs (3.5.12.12).

Add:

63. **Disambiguation of Power Roles (4.7.2.18.13).**

The following test shall be performed. Only the USB Type-C connector will be tested as it is assumed that STUB will be connected to another STUB or DRP device through the USB Type-C if disambiguation of power roles through button press is necessary.

In the following tests, it is assumed that a USB Power Delivery analyzer is connected between the two STUBs to monitor the communication.

To test scenario 1, press STUB's button. Within 10 seconds after pressing the button, connect the STUB to another STUB with a USB Type-C cable. It is preferable to make the connection as late as possible to test that the full 10-second window is implemented. Verify that the STUB whose button was pressed is *Source* and that the other STUB is charging. Unplug the STUBs and repeat this test 10 times. If the STUB whose button was pressed is *Source* every time, the odds are very low ($1/2^{10} \approx 0.1\%$ chance) that it was randomly *Source* every time.

To test scenario 2. a., connect two STUBs together with a USB Type-C cable. Then, press the button on the STUB that is being charged, which can be determined by seeing which STUB's SOCI is blinking. Verify that the *Power Role*s have been swapped by seeing if the STUB's SOCI that was blinking has stopped and the other STUB's SOCI is blinking. Press the button on the STUB that is now charging in order to test the other STUB. Verify that the *Power Role*s have swapped again back to their original roles.

To test scenario 2. b., connect two STUBs together with a USB Type-C cable. Press the button on the STUB that is *Source*. Verify that its SOCI is activated and that nothing else happened. Also verify that the power provided by the *Source* STUB was not interrupted at any time.

The battery shall meet the requirements of Disambiguation of Power Roles (3.5.12.13).

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64. Explosive atmosphere (4.7.3.17).
Operation of the STUB shall consist of the following:
    a. insert a USB-C cable (in Source and Sink modes) into the battery, and then negotiate 20 V Fixed PDO
    b. remove a USB-C cable (in Source and Sink modes) from the battery at the 20 V Fixed PDO
    c. mating the contacts (in Source and Sink modes) in the battery, and then negotiate the 20 V Fixed PDO
    d. unmating the contacts (in Source and Sink modes) in the battery at the 20 V Fixed PDO
    e. activating the SOCI

65. Combined temperature, altitude and humidity (4.7.3.18).
Replace TABLE XVIII with:

TABLE XVIII. *Combined temperature, altitude, and humidity.*

Step	Test	Temperature (°C/°F)	Relative Humidity (%)	Altitude (meters/feet)	Duration (minutes)
1	Ramp to Cold/Dry	-32.0/-25.6	Site	Site	30
2	Cold/Dry Soak	-32.0/-25.6	Uncontrolled	Site	240
4	Ramp to Cold/Dry Altitude	-32.0/-25.6	Uncontrolled	9754/32000	15
5	Cold/Dry Altitude Soak	-32.0/-25.6	Uncontrolled	9754/32000	30
6	Ramp to Warm	32.0/90.0	Uncontrolled	Site	15
7	Ramp to Humid	32.0/90.0	95	Site	15
8	Warm/Humid Soak	32.0/90.0	95	Site	45
9	Hot/Dry Ramp	52.0/125.6	<30	Site	20
10	Hot/Dry Stabilization	52.0/125.6	<30	Site	60
11	Hot/Dry Soak	52.0/125.6	<30	Site	120
13	Ramp to Altitude	52.0/125.6	<30	9754/32000	15
14	Hot/Dry/Altitude Soak	52.0/125.6	<30	9754/32000	240
15	Ramp to Ambient	25.0/77.0	Site	Site	20
16	Ambient Soak	25.0/77.0	Site	Site	30

66. Nail penetration, cell (4.7.4.4).
SOC not less than 50%

67. Overcharge/electrical leakage (4.7.4.7).
    c. Negotiate 20 V *Sink PDO*. After one minute ± 30 seconds, apply a constant 24 V charge voltage with a 4 A limit to *VBUS* on the USB-C interface.
    f. Discharge according to the standard rate in TABLE iv.
    k. Additional charge contacts are E1, E2, and E3 (see FIGURE 2).

68. Projectile (4.7.4.11).
SOC not less than 50%

69. Nail penetration, battery (4.7.4.12).
SOC not less than 50%

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70. End item interoperability (4.7.6.2).
Verify the end item(s) remains fully functional in accordance with all modes of operation with the battery installed. Verify power flow is in the correct direction for each mode of operation. Additional charge contacts are E1, E2, and E3 (see FIGURE 2).

71. Electromagnetic interference (4.7.7).
The battery EMI shall be characterized. Testing shall be conducted in accordance with associated test procedures of MIL-STD-461. Conducted tests shall reflect battery contributions to MIL-STD-461 requirements and shall be recorded and verified to not exceed that of MIL-STD-461 limits.
Note, the Government may require personnel with security clearance or may conduct independent testing and adjudication of independent test results.

The following configuration for Electromagnetic Interference (EMI) Testing shall be used:

a. Configuration 1 (ref: MIL-STD-461, FIGURE 2) has the battery under test attached to a USB-PD capable unit(s) with *Source* and/or *Sink* capabilities. Discharge batteries, through the contacts, at the high rate *PDO* and rate per TABLE iii and TABLE iv, respectively. Charge batteries, through the contacts, in a *Fixed PDO* at a rate not less than offset byte 20 of the SKEDB per 3.5.12.4.

CE106, conducted emissions, antenna port: N/A

CS103, conducted susceptibility, antenna port, intermodulation: N/A

CS104, conducted susceptibility, antenna port, rejection of undesired signals: N/A

CS105, conducted susceptibility, antenna port, cross modulation: N/A

CS109, conducted susceptibility, structure current: N/A

CS114, conducted susceptibility, bulk cable injection: Testing shall utilize AIRCRAFT (EXTERNAL OR SAFETY CRITICAL) level limits CS114 limit curves (ref: MIL-STD-461, TABLE VI) for Army platforms. The injection and monitor probes shall be placed on the USB Type-C cable bundle including signal and ground, 5 cm from the equipment under test (EUT). Verify by testing using Configuration 1 that the battery complies with FIGURE CS114-1 limits.

CS115, conducted susceptibility, bulk cable injection, impulse excitation: Verify by test using Configuration 1 that the battery exhibits no performance degradation.

CS116, conducted susceptibility, damped sinusoidal transients, cables and power leads: Use limits of FIGURE CS116-2. Verify by test using Configuration 1 that the battery complies with FIGURE CS116-2 limits.

CS117, conducted susceptibility, lightning induced transients, cables and power leads: N/A

CS118, personnel borne electrostatic discharge: Test points shall include, at a minimum,
Air Method: Battery housing faces, seams and openings
Contact Method: Battery contacts.
Verify by test using Configuration 1 that the battery exhibits no performance degradation.

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RE101, radiated emissions, magnetic field: Use the limits of FIGURE RE 101-1 ("RE101-1. RE101 limit for all Army applications"). Verify by test using Configuration 1 that the battery complies with FIGURE RE101-1 limits.

RE102, radiated emissions, electric field: Use the limits of FIGURE RE102-4 ("Navy Mobile & Army") limit in FIGURE RE102-4. RE102 limit for ground applications. Verify by test using Configuration 1 that the battery complies with FIGURE RE102-4.

RE103, radiated emissions, antenna spurious and harmonic outputs: N/A

RS101, radiated susceptibility, magnetic field: N/A

RS103, radiated susceptibility, electric field: Testing shall utilize "AIRCRAFT (EXTERNAL OR SAFETY CRITICAL)" level limits for 2 MHz to 18 GHz for Army platforms defined in TABLE XI. RS103 limits. Test to be conducted in Configuration 1. Record thresholds of susceptibility in accordance with MIL-STD-461 paragraph 4.3.10.4.3 and data presentation requirements of 5.21.4.5 and mark all data in accordance with security guidelines.

RS105: N/A

72. Approved chargers (6.6).
Remove:

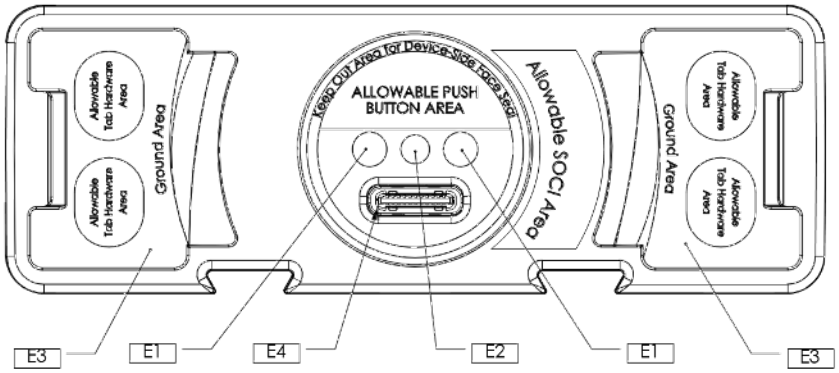
DESIGNATION	COMMON NAME	NIIN	PART NUMBER	CAGE
PP-8498/U	Soldier Portable Charger (SPC)	01-495-2839	BTC-70801	51828
PP-8481B/U	Vehicle Mounted Charger (VMC)	01-527-2726	BTC-70836	51828

ADDITIONAL FIGURES
(On the following pages)

Note: Dimensions are in inches. Tolerances: if one decimal is present, 0 ± 0.1 inches, if two decimals are present, 0 ± 0.01 inches, if three decimals are present, 0 ± 0.005 inches, and if an angle, 0 ± 1°, unless stated otherwise.

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ELECTRICAL CONNECTIONS



CONTACT REF DES	CONNECTION
E1	VBUS
E2	CC
E3	GND
E4	USB-C

NOTES: (UNLESS OTHERWISE SPECIFIED)

1. VBUS HAS NO OUTPUT UNTIL CC CONNECTION IS PRESENT PER USB POWER DELIVERY SPECIFICATION REVISION 3.2 VERSION 1.1 SECTION 2.6.4.1
2. STATE OF CHARGE INDICATOR (SOCI) SHALL FILL SEQUENTIALLY FROM DOVETAIL SIDE TO OPPOSING SIDE

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: ANGULAR: MACH ± 1° TWO PLACE DECIMAL ±.01 THREE PLACE DECIMAL ±.005 INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5 THIRD ANGLE PROJECTION DO NOT SCALE DRAWING	TITLE: STUB 2.0 INTERFACE CONTROL DOCUMENT SIZE B REV A SCALE: N/A

FIGURE 2. BB-25XXA/U electrical connections

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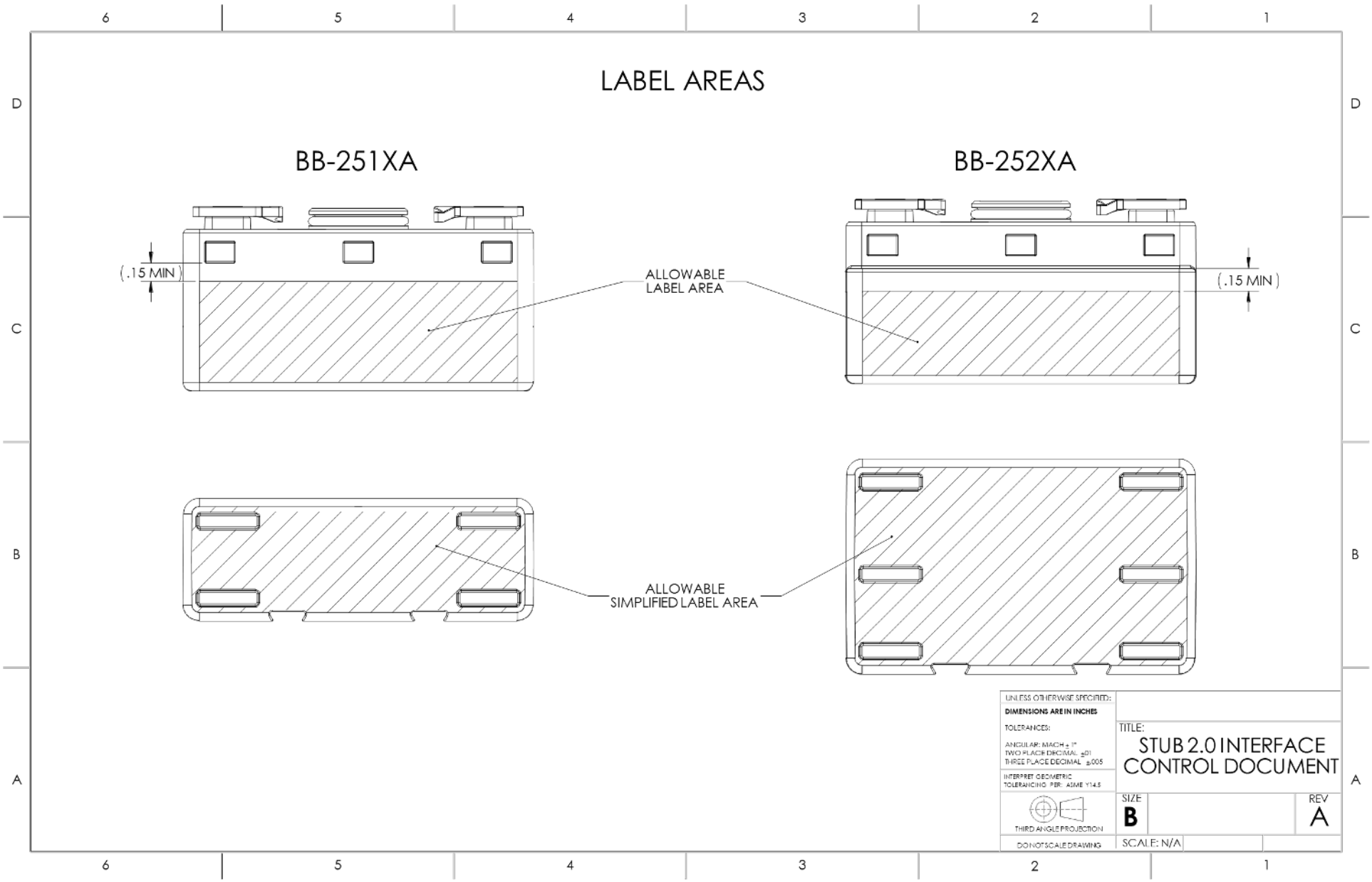


FIGURE 3. BB-251XA/U and BB-252XA/U allowable label areas

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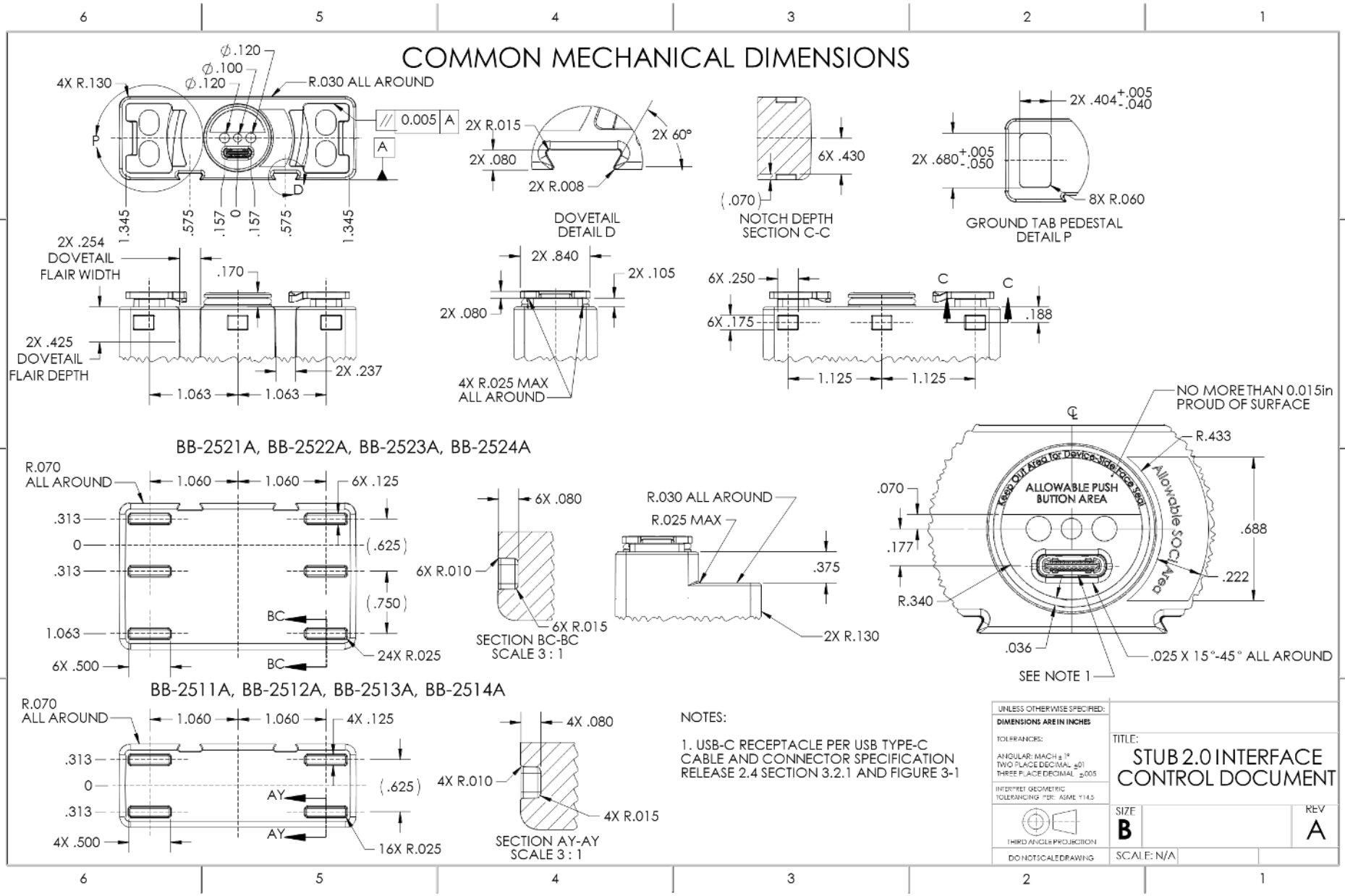


FIGURE 4. BB-25XXA/U common mechanical dimensions

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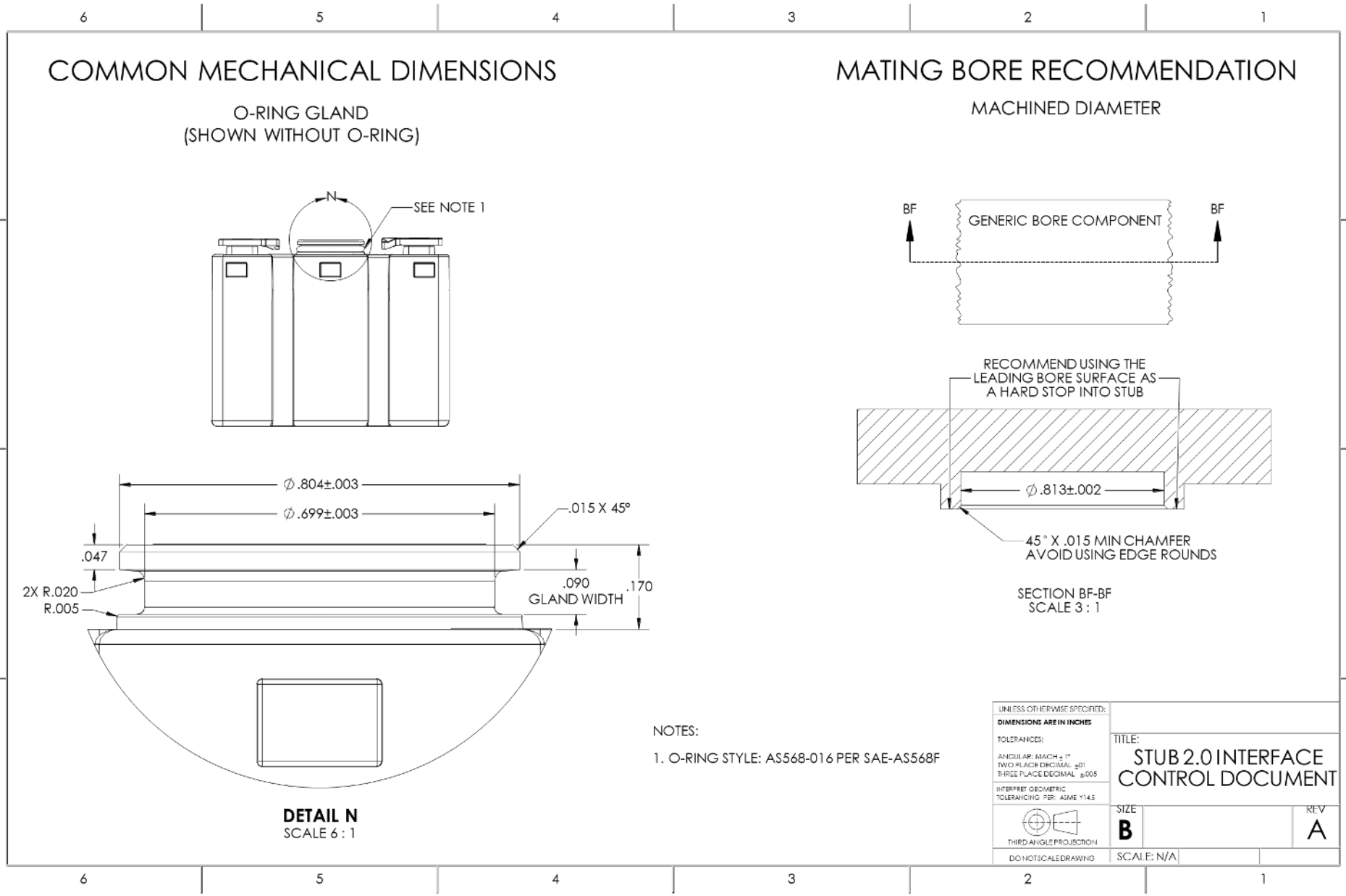


FIGURE 5. BB-25XXA/U O-ring gland

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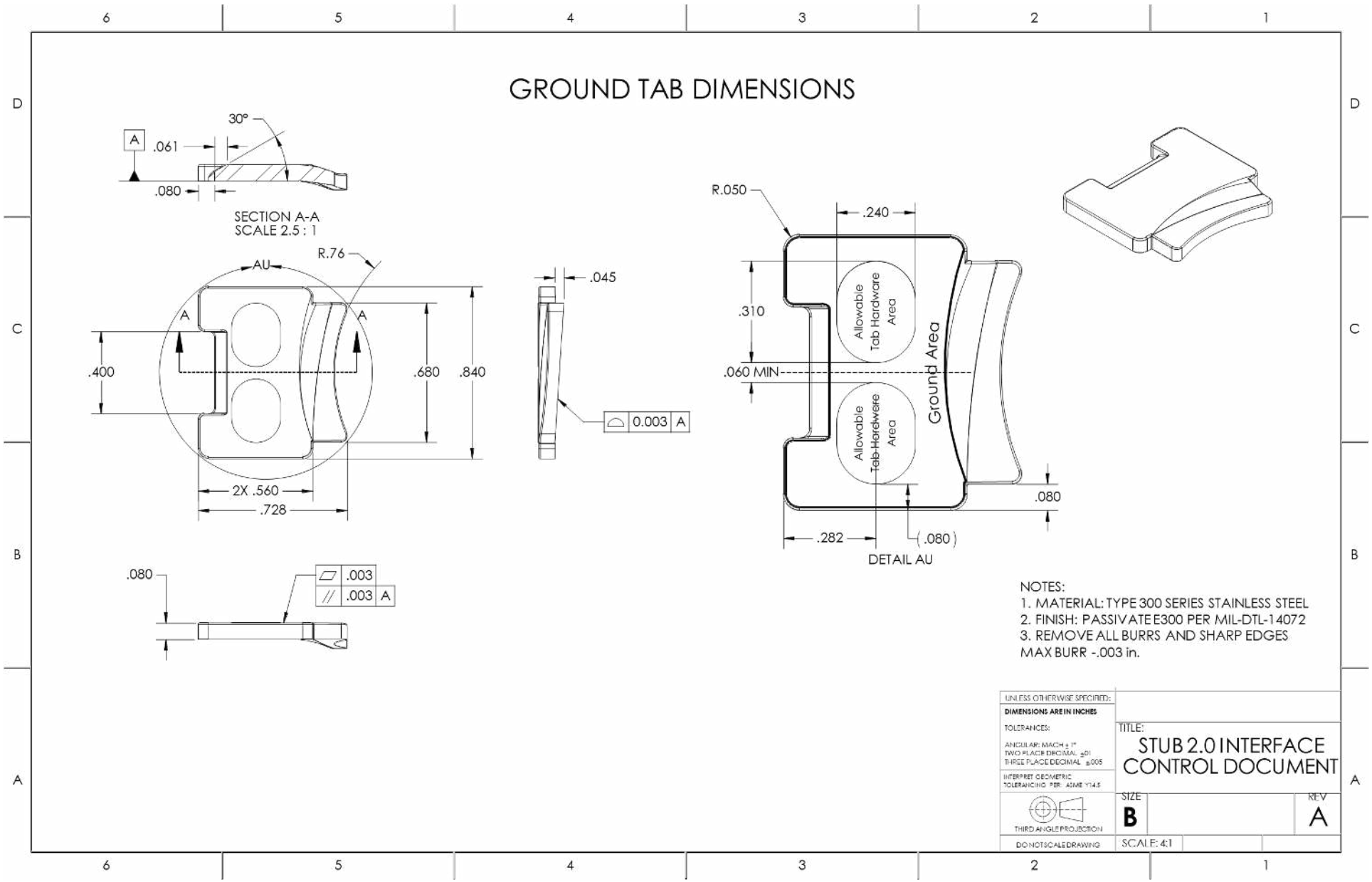


FIGURE 6. BB-25XXA/U ground tab

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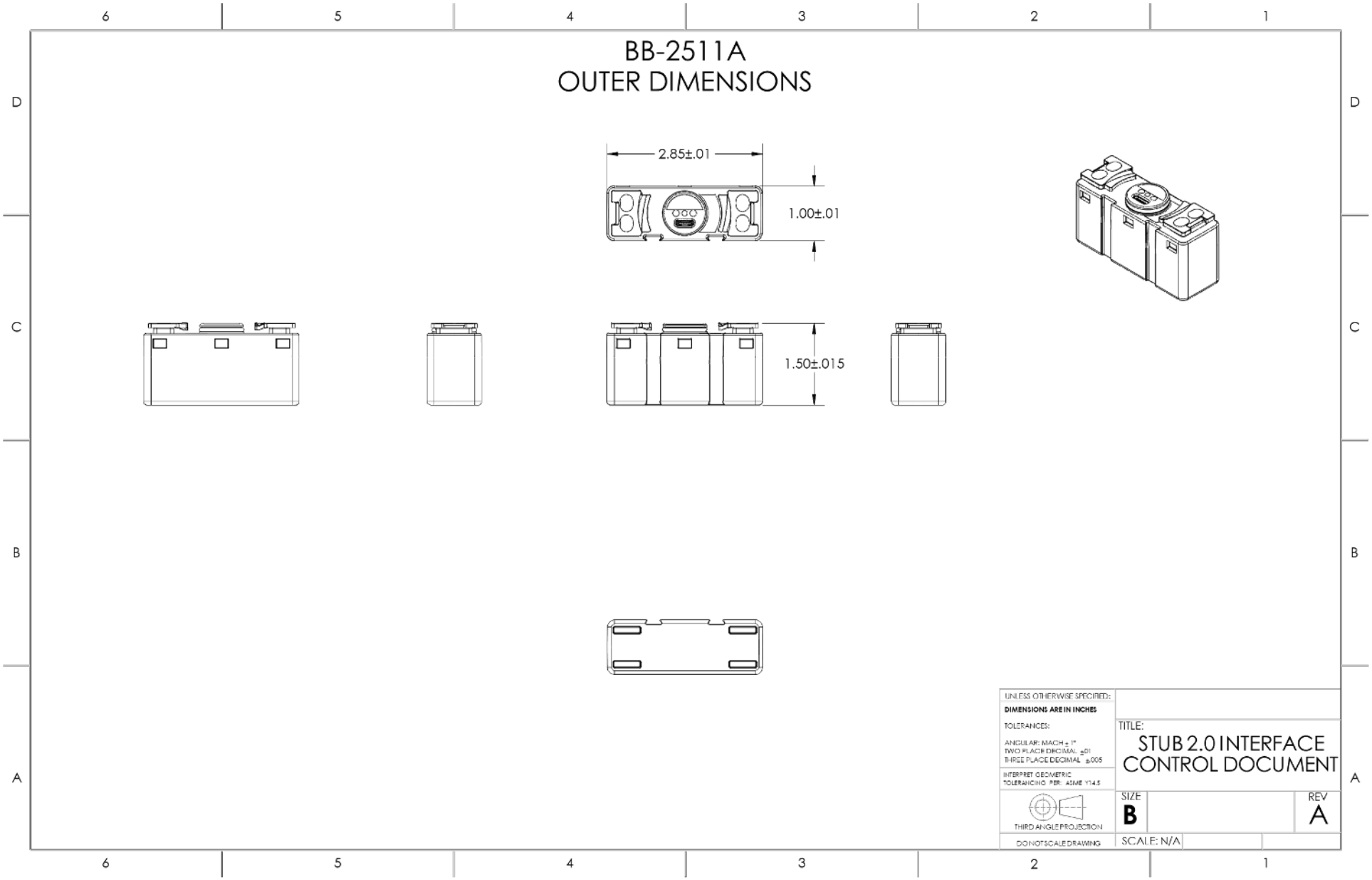


FIGURE 7. BB-2511A/U

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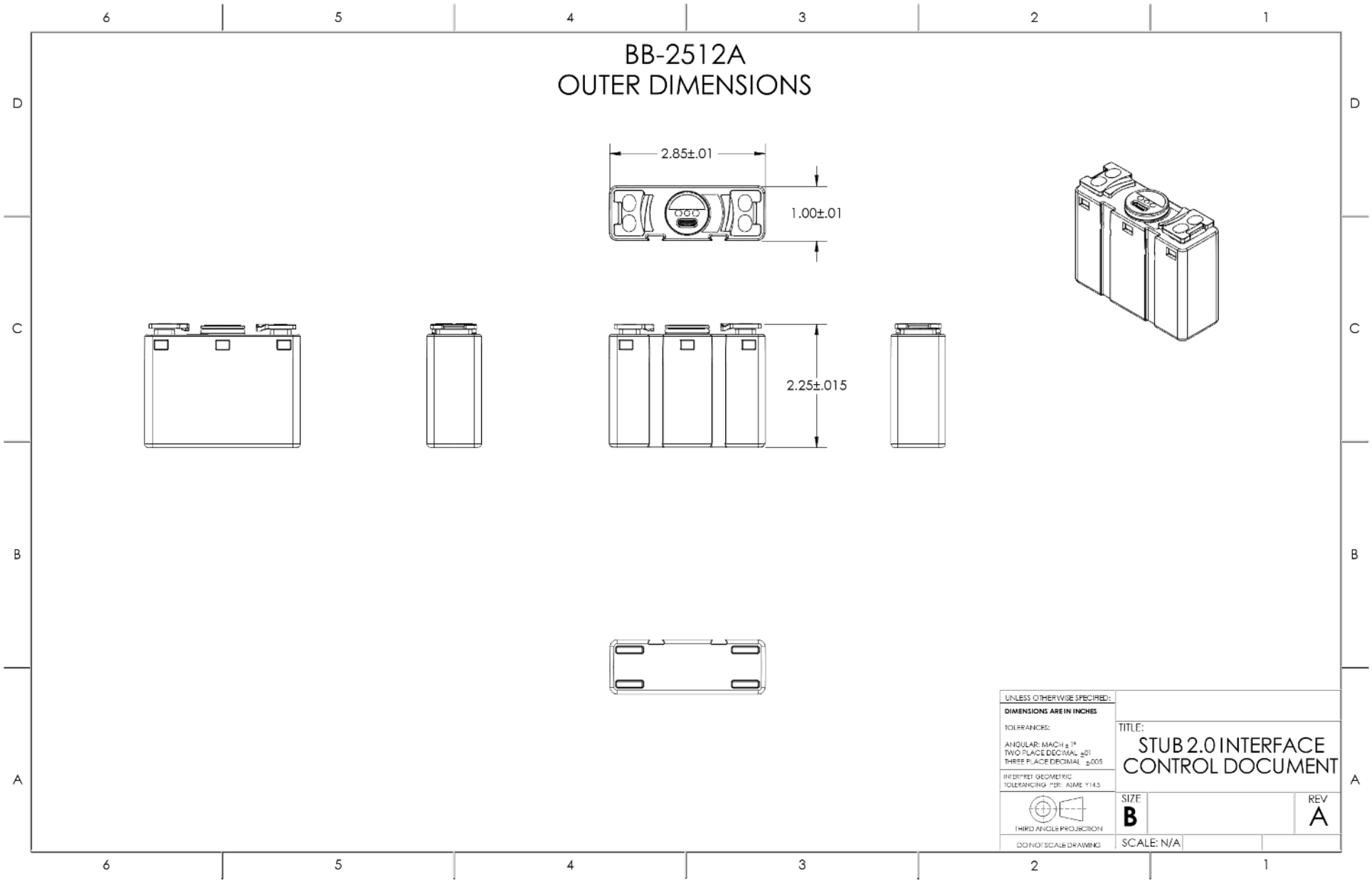


FIGURE 8. BB-2512A/U

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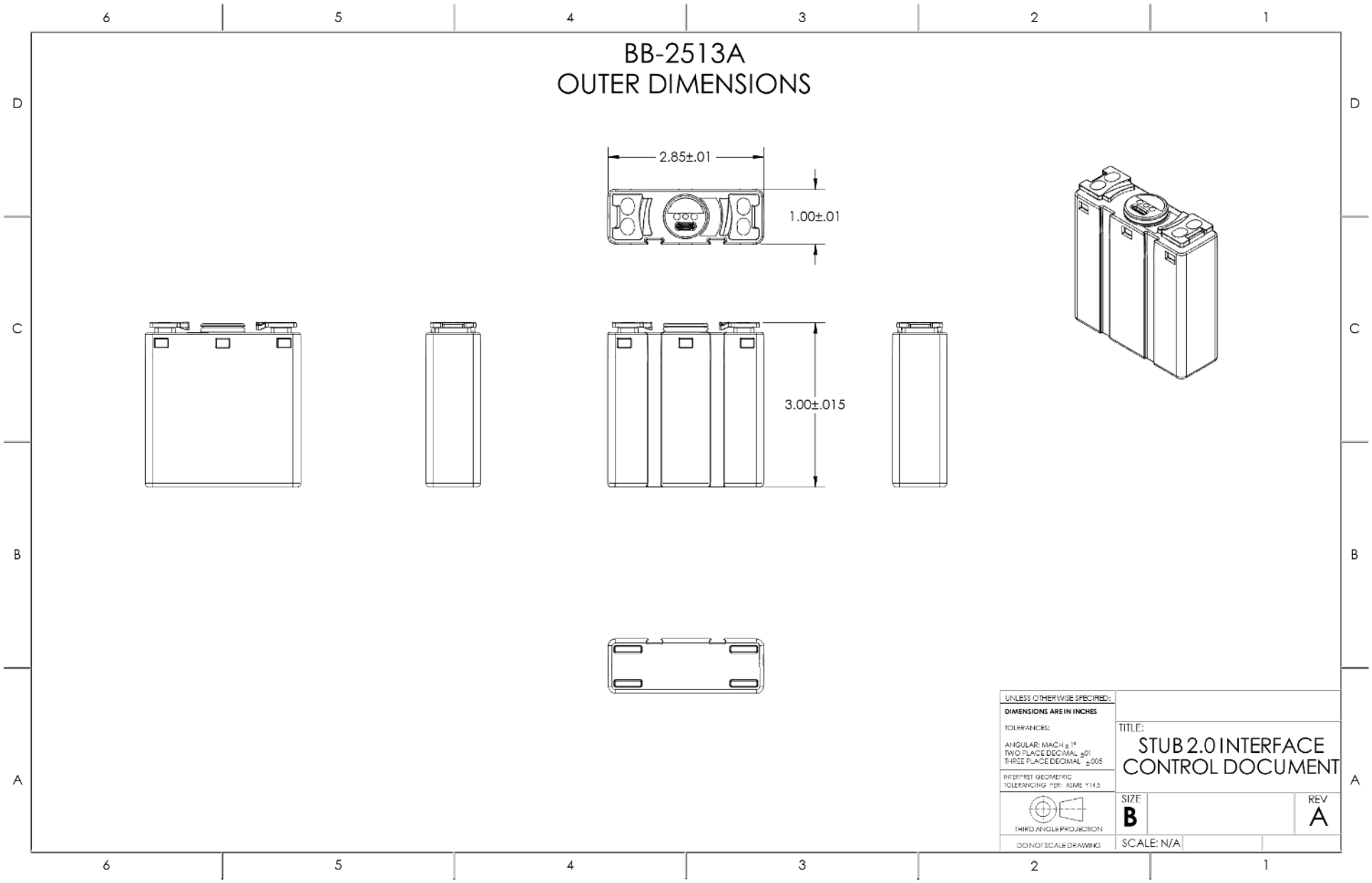


FIGURE 9. BB-2513A/U

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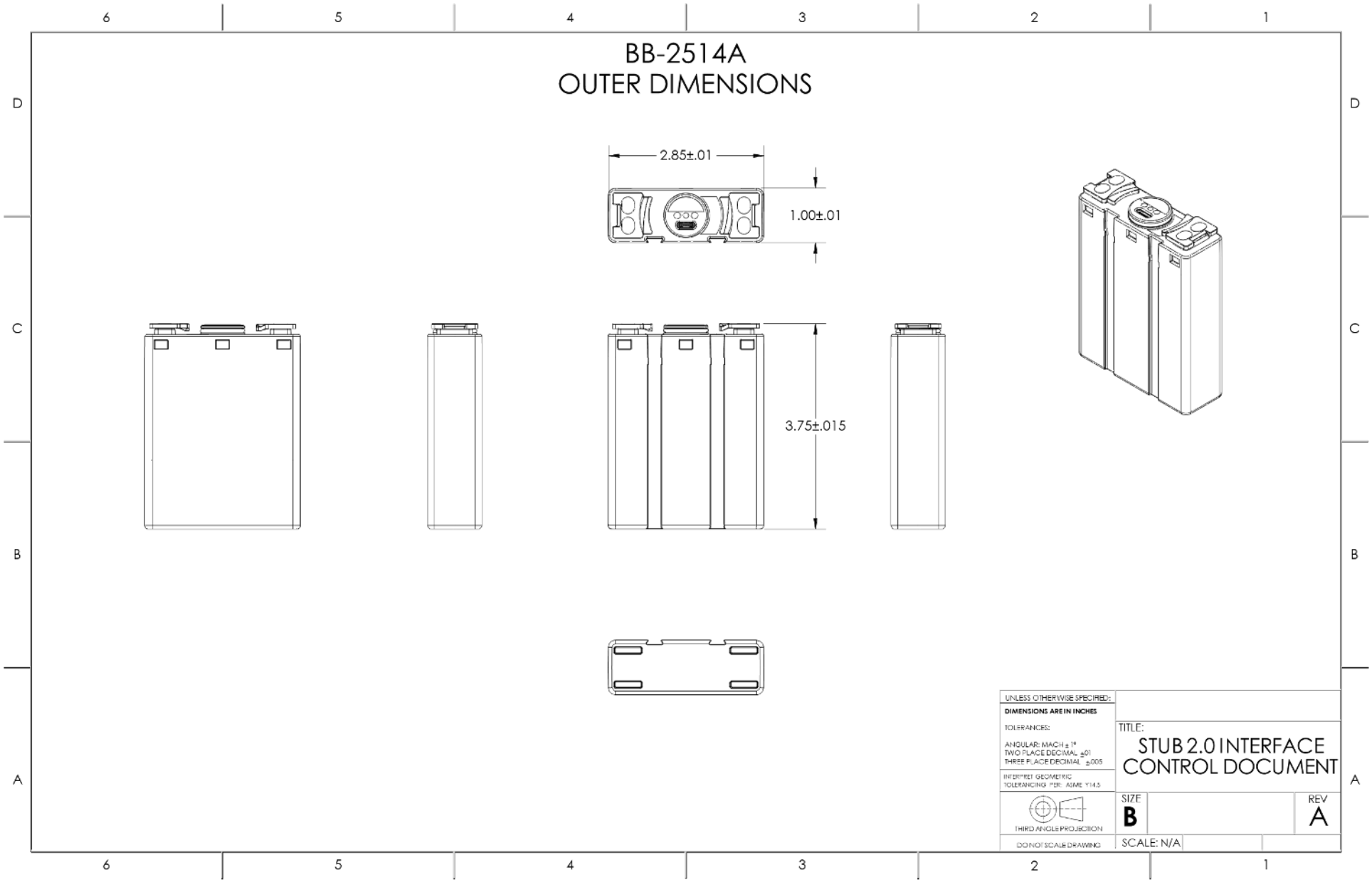


FIGURE 10. BB-2514A/U

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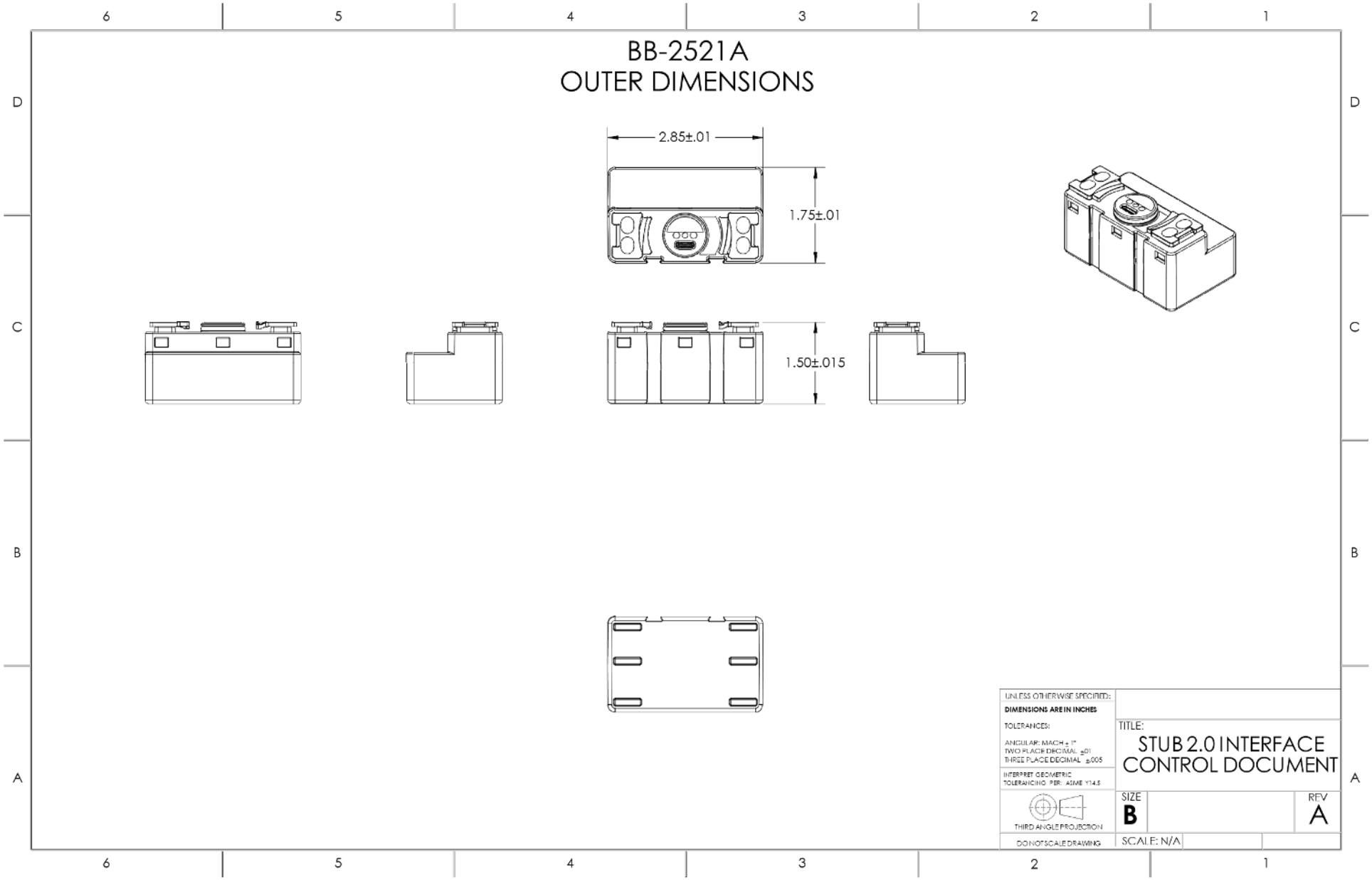


FIGURE 11. BB-2521A/U

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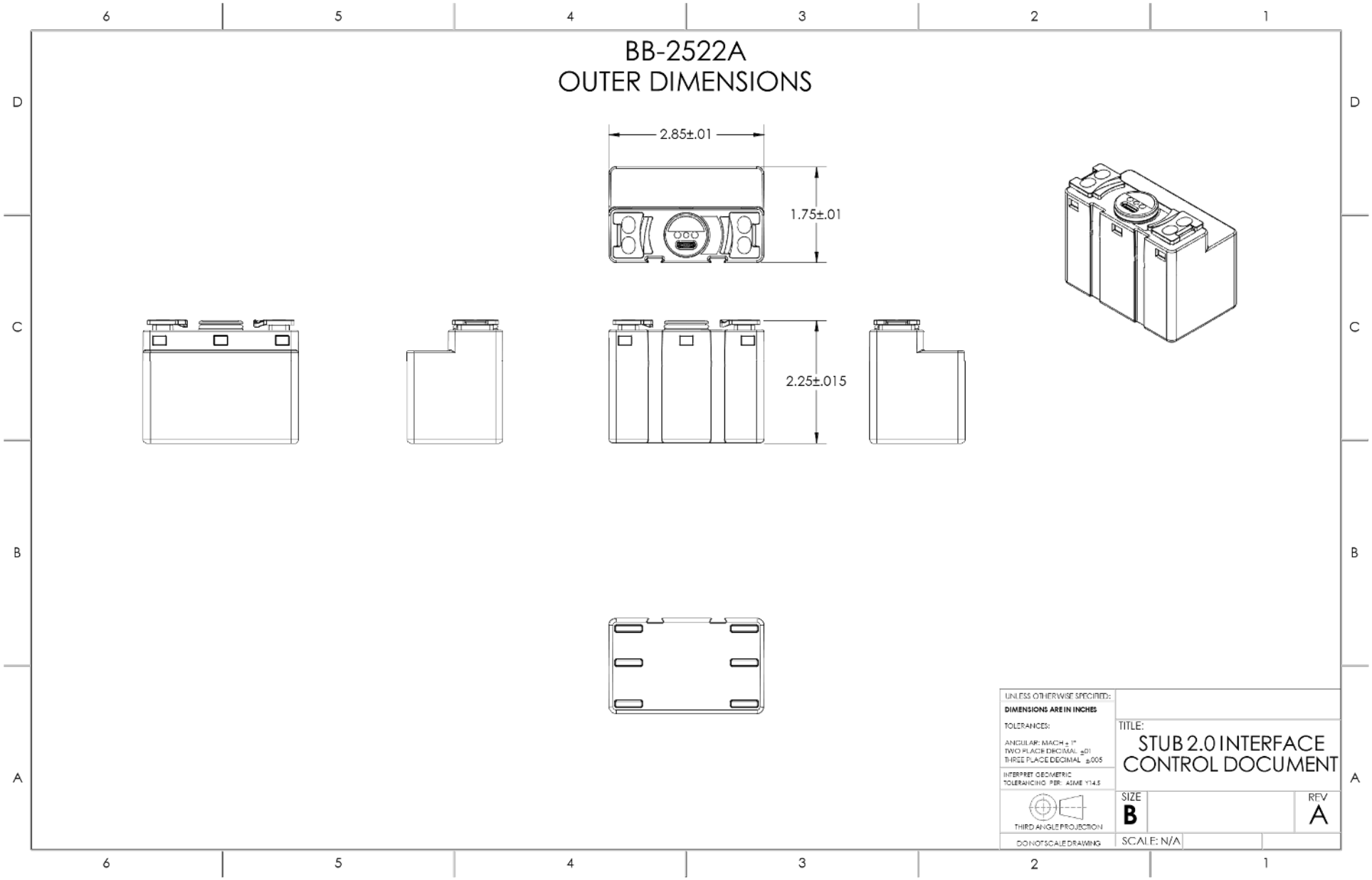


FIGURE 12. BB-2522A/U

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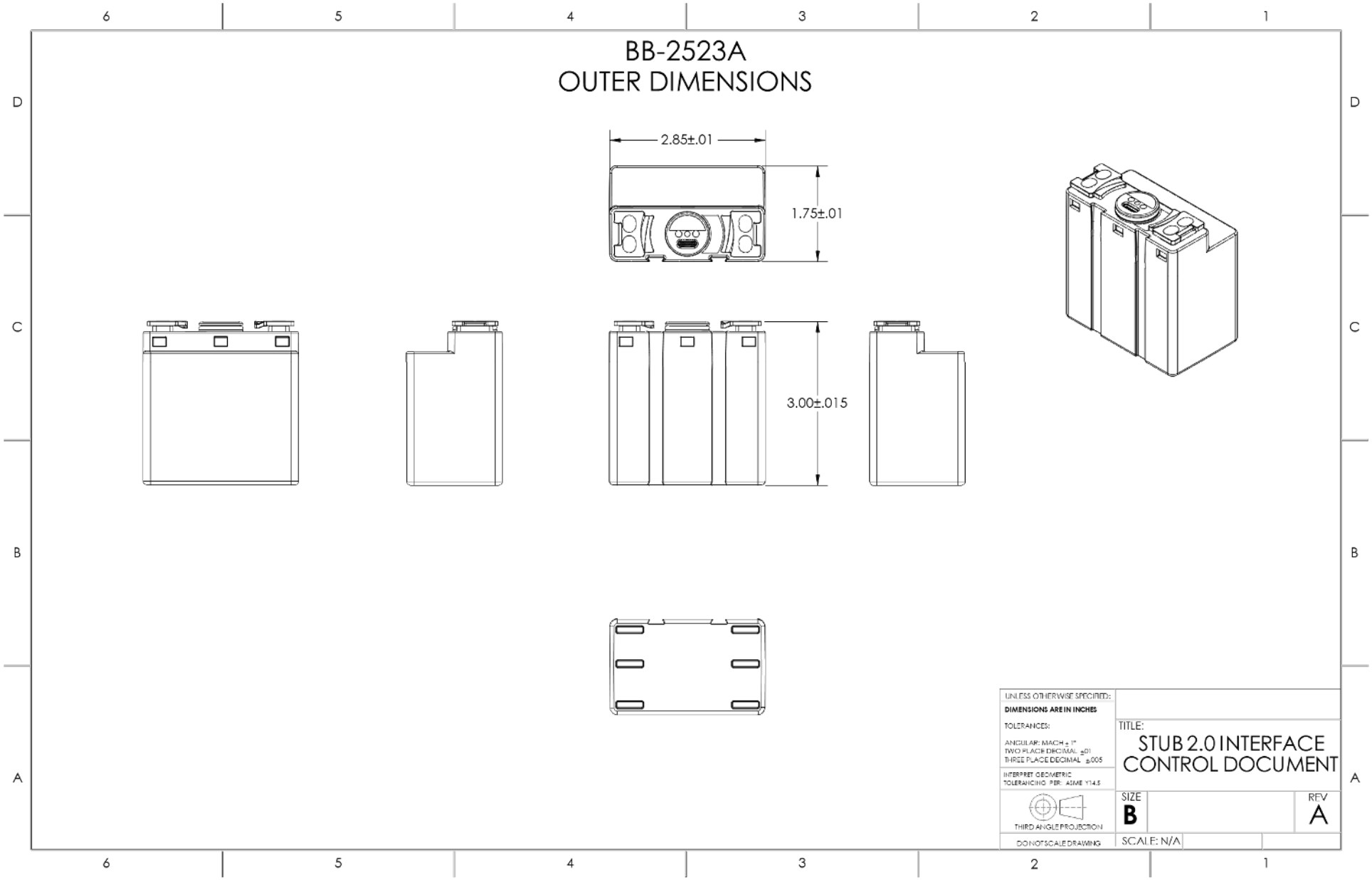


FIGURE 13. BB-2523A/U

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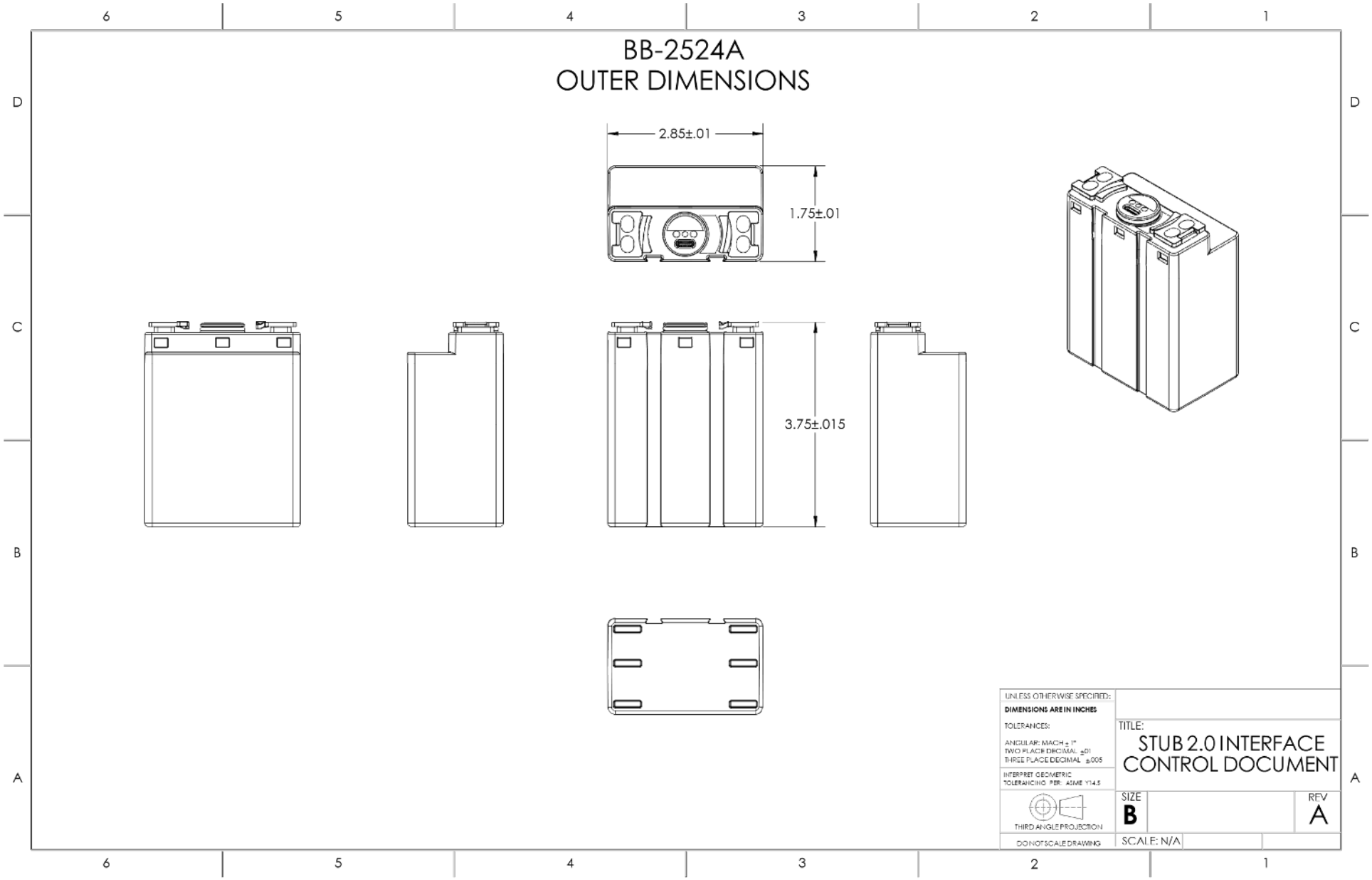


FIGURE 14. BB-2524A/U

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NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Nominal ratings: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

TABLE ix. *STUB nominal ratings*

Battery PIN: M32383/7A		**-3-1X**				**-3-2X**			
Type Classification:		BB-2511A	BB-2512A	BB-2513A	BB-2514A	BB-2521A	BB-2522A	BB-2523A	BB-2524A
Chemistry:		Li-Ion							
Color:		Tan 686A							
Overall Dimensions (inches):		2.85 x 1.00 x 1.50	2.85 x 1.00 x 2.25	2.85 x 1.00 x 3.00	2.85 x 1.00 x 3.75	2.85 x 1.75 x 1.50	2.85 x 1.75 x 2.25	2.85 x 1.75 x 3.00	2.85 x 1.75 x 3.75
Weight (max):		0.30 lbs	0.40 lbs	0.55 lbs	0.65 lbs	0.40 lbs	0.65 lbs	0.90 lbs	1.20 lbs
Voltage Range:		5-20 V							
Energy (new battery):		11 Wh	22.4 Wh	34 Wh	45.8 Wh	22.4 Wh	45.8 Wh	70 Wh	94 Wh
SOURCE PDP rating:		18 W	36 W	54 W	72 W	36 W	72 W	100 W	100 W
SOURCE Discharge Levels:	*Standard*	5V / 2.75W	9V / 5.6W	9V / 8.5W	15V / 11.45W	9V / 5.6W	15V / 11.45W	20V / 17.5W	20V / 23.5W
	High Rate	5V / 9W	9V / 18W	9V / 27W	15V / 36W	9V / 18W	15V / 36W	20V / 50W	20V / 50W
	Pulse	Standard for 144s, then 1.5 x Standard for 18s, then PDP for 18s							
	Peak	150% PDP for 1ms at 5% duty cycle, 125% PDP for 2ms at 10% duty cycle, 110% PDP for 10ms at 50% duty cycle							
Available SOURCE Power Data Objects (PDOs):	*Contacts* *Fixed*	5V / 3.60A 9V / 2.00A 15V / 1.20A 20V / 0.90A	5V / 5.00A 9V / 4.00A 15V / 2.40A 20V / 1.80A	5V / 5.00A 9V / 5.00A 15V / 3.60A 20V / 2.70A	5V / 5.00A 9V / 5.00A 15V / 4.80A 20V / 3.60A	5V / 5.00A 9V / 4.00A 15V / 2.40A 20V / 1.80A	5V / 5.00A 9V / 5.00A 15V / 4.80A 20V / 3.60A	5V / 5.00A 9V / 5.00A 15V / 5.00A 20V / 5.00A	5V / 5.00A 9V / 5.00A 15V / 5.00A 20V / 5.00A
	SPR AVS	9-15V / 1.20A 15-20V / 0.90A	9-15V / 2.40A 15-20V / 1.80A	9-15V / 3.60A 15-20V / 2.70A	9-15V / 4.80A 15-20V / 3.60A	9-15V / 2.40A 15-20V / 1.80A	9-15V / 4.80A 15-20V / 3.60A	9-15V / 5.00A 15-20V / 5.00A	9-15V / 5.00A 15-20V / 5.00A
	SPR PPS	5-11V / 2.00A	5-11V / 4.00A	5-16V / 3.60A	5-21V / 3.60A	5-11V / 4.00A	5-21V / 3.60A	5-21V / 5.00A	5-21V / 5.00A
	Battery	2.5-4.2V / 18W	5.0-8.4V / 36W	7.5-12.6V / 54W	10-16.8V / 72W	5.0-8.4V / 36W	10-16.8V / 72W	7.5-12.6V / 60W	10-16.8V / 80W
	USB-C *Fixed*	5V / 3.60A 9V / 2.00A N/A N/A	5V / 5.00A 9V / 4.00A 15V / 2.40A N/A	5V / 5.00A 9V / 5.00A 15V / 3.60A 20V / 2.70A	5V / 5.00A 9V / 5.00A 15V / 4.80A 20V / 3.60A	5V / 5.00A 9V / 4.00A 15V / 2.40A N/A	5V / 5.00A 9V / 5.00A 15V / 4.80A 20V / 3.60A	5V / 5.00A 9V / 5.00A 15V / 5.00A 20V / 5.00A	5V / 5.00A 9V / 5.00A 15V / 5.00A 20V / 5.00A
	SPR AVS	N/A N/A	9-15V / 2.40A N/A	9-15V / 3.60A 15-20V / 2.70A	9-15V / 4.80A 15-20V / 3.60A	9-15V / 2.40A N/A	9-15V / 4.80A 15-20V / 3.60A	9-15V / 5.00A 15-20V / 5.00A	9-15V / 5.00A 15-20V / 5.00A
	SPR PPS	5-11V / 2.00A	5-11V / 4.00A	5-16V / 3.60A	5-21V / 3.60A	5-11V / 4.00A	5-21V / 3.60A	5-21V / 5.00A	5-21V / 5.00A
	Battery	2.5-4.2V / 12.5W	5.0-8.4V / 25W	7.5-12.6V / 37.5W	10-16.8V / 50W	5.0-8.4V / 25W	10-16.8V / 50W	7.5-12.6V / 37.5W	10-16.8V / 50W
Battery Life:		≥ 224 cycles, ≥ 3 years							
Temperature ranges:		Charge: -4 to 122ºF (-20 to 50ºC)			Operational: -22 to 131ºF (-30 to 55ºC)			Storage: -4 to 122ºF (-20 to 50ºC)	
		MAX abusive temperature (non-operating): 199ºF (93ºC)							
Interface:		Contacts, USB-C (See FIGURE 2)							

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Other data:
Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for the BB-25XXA/U batteries during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

USB Power Delivery protocol keyword formatting:
The text formatting, including font color, of USB Power Delivery protocol keywords is taken from the USB Power Delivery Specification.

Reason for disambiguation of power roles:
According to Section 4.5.3.1.4 DRP to DRP Behavior in the USB Type-C Cable and Connector Specification, when one Dual-Role Power (DRP) device is connected to another DRP device, “the two DRPs accept the resulting Source-to-Sink relationship achieved randomly”. Because STUB is a DRP device, when a user connects one STUB to another STUB in an attempt to charge one STUB with the other, it will be random which one is the *Source* and charging the other, which could be confusing to the user. The same randomness would happen if a user connects STUB to another DRP, such as a laptop or a phone. To make STUB-to-STUB and STUB-to-DRP charging straightforward.

STUB nomenclature:
In determining the nomenclatures for these batteries, the government used the following structure:

BB-25XY

BB – Rechargeable
25 – Li-Ion Chemistry
X – Battery width reference number
- 1 – 1.00in
- 2 – 1.75in

Y – Battery height reference number
- 1 – 1.50in
- 2 – 2.25in
- 3 – 3.00in
- 4 – 3.75in

Methods of attaching STUB to a host device:
The image sequences of FIGURE 15 illustrate the intended approaches a host device may be designed to take to attach a STUB.

a. Insertion: Intends to use the STUB’s dovetail slots or a slot in the device housing, plus either any of the housing indents or under tab retention
b. Hook: Intends to use a fixed and a movable ledge for under tab retention
c. Slide: Intends to rails for under tab retention, plus tab cutouts
d. Twist: Intends to use the inclined planes of the tabs, plus tab cutouts

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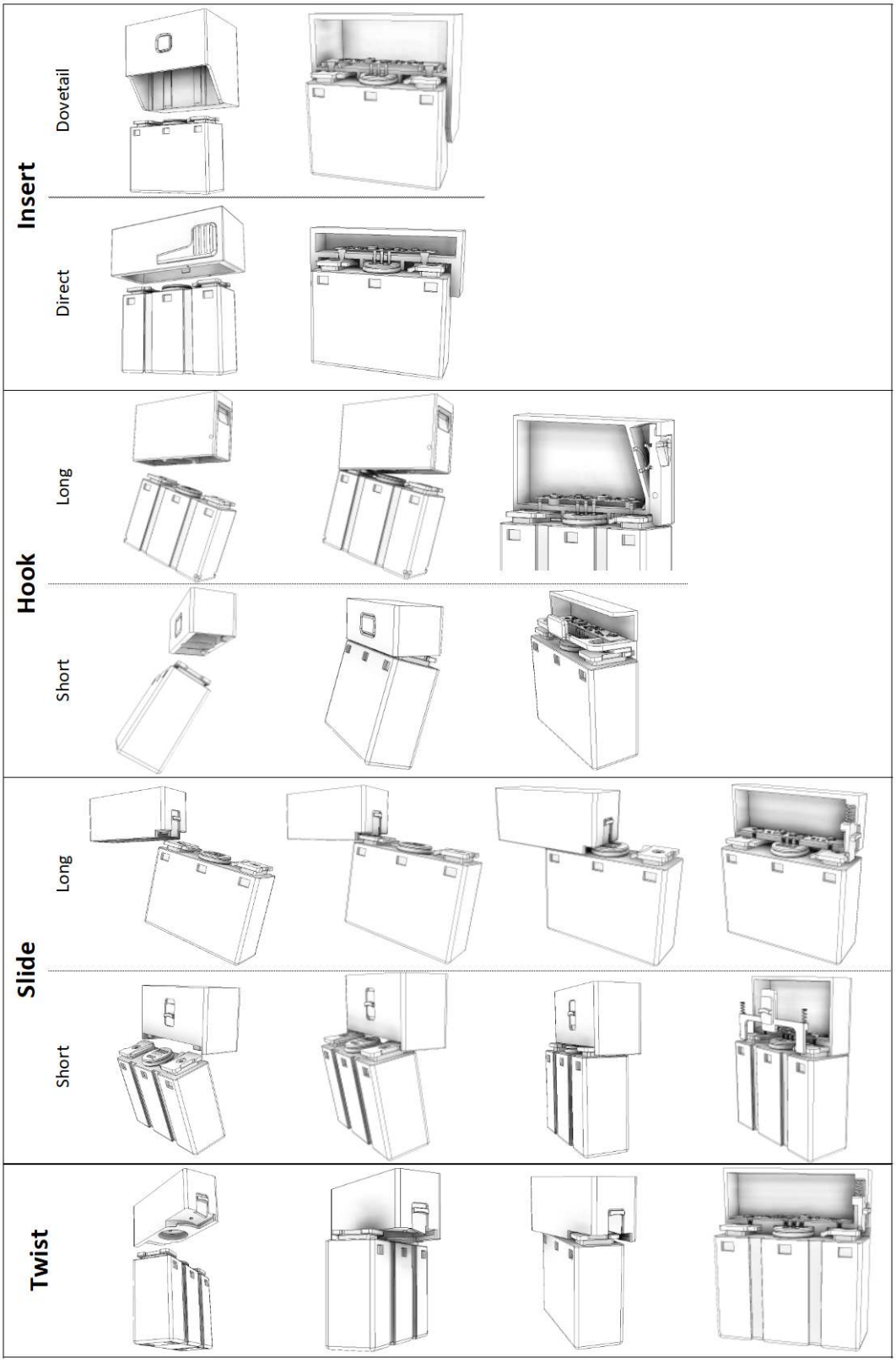


FIGURE 15. Methods of attachment

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USB logo display requirements.
Per USB-IF, to qualify for the right to display the certified USB logo in conjunction with a product, the product must pass USB-IF compliance testing for product quality. It is the vendor's responsibility to meet the requirements as defined in the USB-IF Trademark License Agreement Mark Requirements Chart.

Changes from previous issue.
Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
DLA – CC

Preparing Activity
Army – CR
(Project Number 6140-2026-012)

Review activities:
Navy – AS, SH
Air Force – 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil or https://quicksearch.dla.mil.

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